Lab #2: DC-DC boost prep

Name: Tony Han

Table partners: Marnu, Pranay, Theo

1. Experiment Description and Prelab

1.1: Description

In this lab, we will explore some circuit behaviour to prepare us for building a DC-DC boost circuit. Here, we will conduct experiments involving RC circuits with time-dependent behaviour. This includes: using an RC circuit to measure resistance and capacitance values for an unknown resistor and capacitor, measure the 3dB point of an RC high pass filter and low pass filter, then use this to design a filter to clean the noise out of a given sinusoidal wave. Note: we will have to design a circuit to measure the unknown values of capacitance and resistance, rather than using the volt meter!!

##1.2 Prelab activities

1. Estimate of the time constant for the circuit:

```
In [6]: InlineImage('prelabdrawing.png', width=13, height=8, caption= "Figure 1, circuit diagram and time constant calculations", fs=24);
```

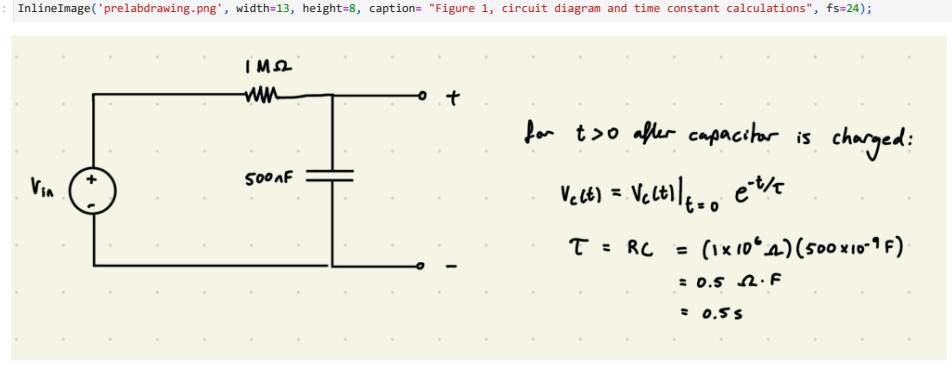

1 MΩ
500nF
Vin
+
−
for t>0 after capacitor is charged:
Vc(t) = Vc(t)|t=0 e^(−t/τ)
τ = RC = (1×10⁶ Ω)(500×10⁻⁹ F)
= 0.5 Ω·F
= 0.5s

Figure 1, circuit diagram and time constant calculations

The approximate time constant for this circuit in the prelab is 0.5s.

Now, for $V_c(t=0)$ if $V_{in}(t<0)=5V$: We can explain this with logic. Assuming the capacitor gets fully charged, there will be no current in the circuit anymore, and the capacitor acts as an open circuit. Consequently, the resistor will have no voltage drops through it as the voltage drop through a resistor depends on the current flowing through it. As such, in order for the circuit to obey the laws of conservation of charge, the voltage drop in the charged resistor. However, if we don't know that the capacitor has been fully charged, the equation for $V_c(t=0)$ will have to depend on the equation for a charging capacitor, which will be a lot more complicated

2. Graphs for the expected waveforms of $V_{in}(t)$ and $V_c(t)$ if I have a square wave as the source, and the circuit is connected:

```
In [16]: InlineImage('squarewavess.png', width=13, height=16, caption= "Figure 2, waveform predictions for Vin and Vc", fs=24);
```

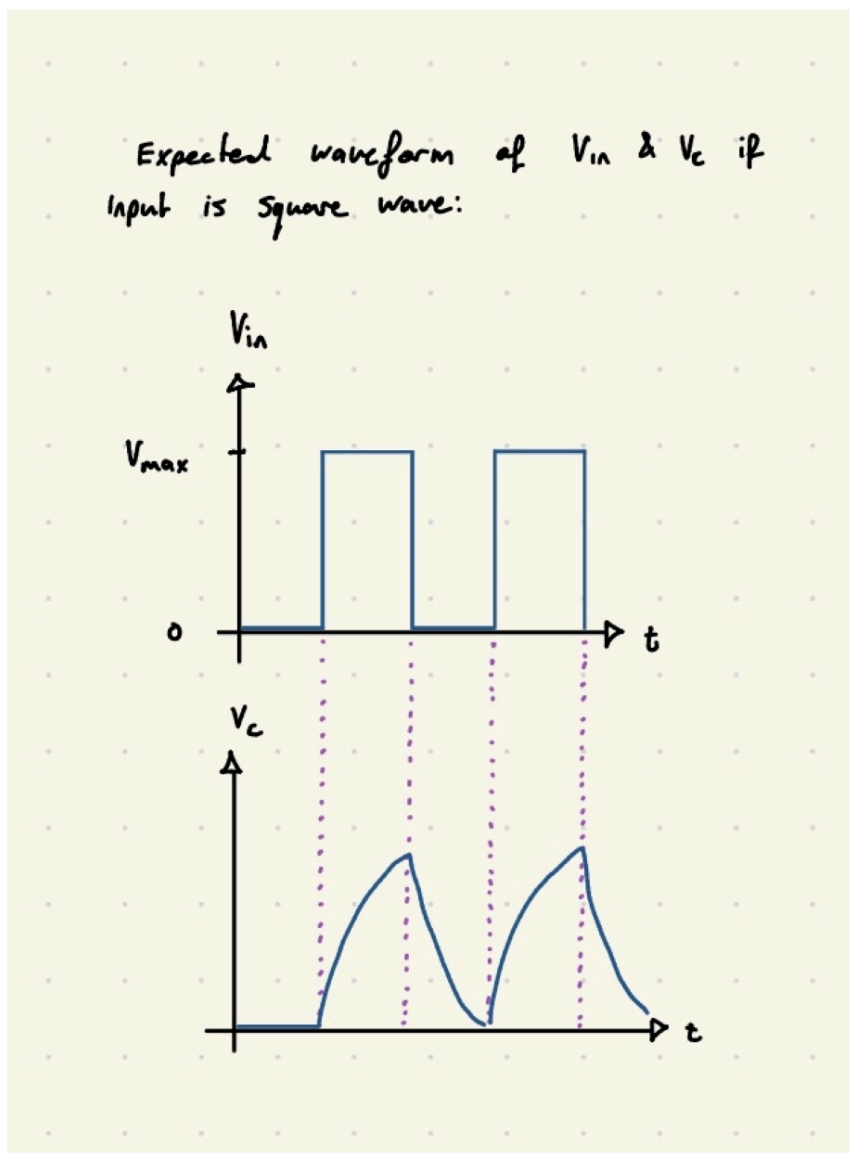

Expected waveform of Vin & Vc if
Input is Square wave:
Vin
Vmax
0
t
Vc
t

Figure 2, waveform predictions for Vin and Vc

For clearer communication, I have drawn the graphs on top of each other with lines corresponding to key moments in time. I have also not labelled the voltage on the y axis for V_c as it can change depending on the frequency of the square wave. It can be anywhere from 1% to 100% of the maximum stored voltage on the capacitor. So, I have left it blank to show that the peaks of the wave can be a multitude of values, depending on the frequency of the source square wave.

To measure the time constant of the circuit, we need to choose a frequency such that the whole waveform of the capacitor is shown. In essence: we need to allow enough time for the capacitor to fully charge and discharge. To know how long this is, we should look at the time constants: Given a τ of 0.5s (as calculated in figure 1), we know that the capcitor will discharge approx. 63% per 0.5s. Since this is not a linear decay, we can't simply take 2 τ as the period. I will calculate how many τ we need the period to be in the figure below. From then, we can take the inverse to find our desired frequency.

```
In [15]: InlineImage('taucac.png', width=20, height=8, caption= "Figure 3, Calculations for period", fs=24);
```

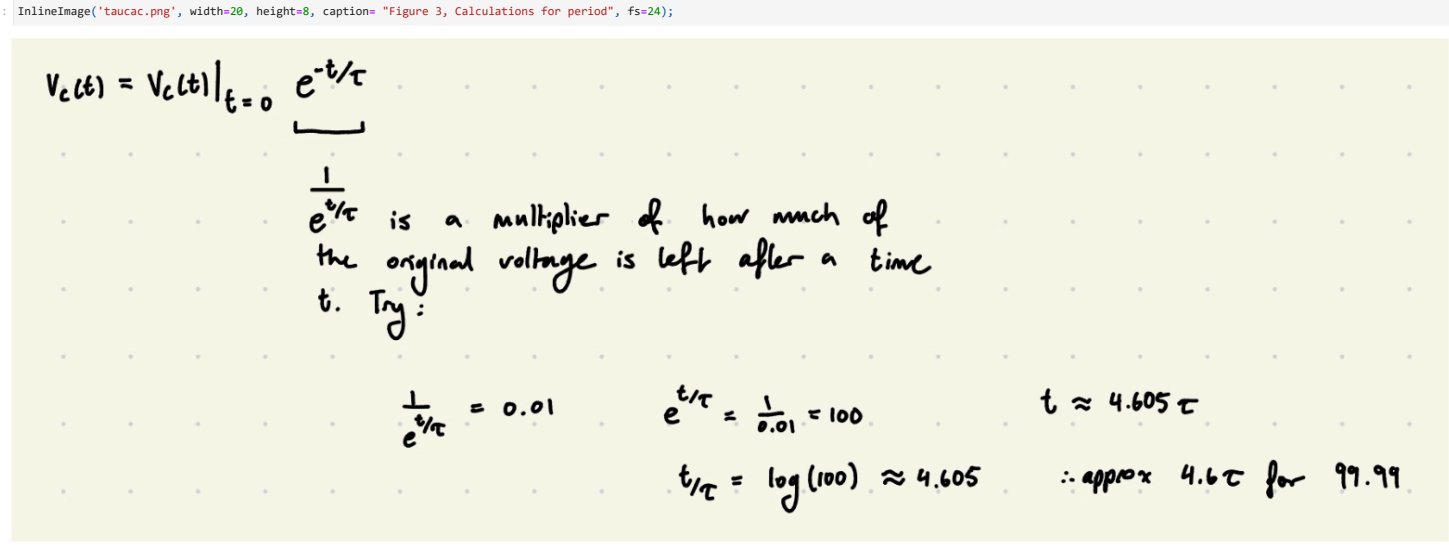

Vc(t) = Vc(t)|t=0 e^(−t/τ)
1/e^(t/τ) is a multiplier of how much of the original voltage is left after a time t. Try:
1/e^(t/τ) = 0.01
e^(t/τ) = 1/0.01 = 100
t/τ = log(100) ≈ 4.605
t ≈ 4.605 τ
∴ approx 4.6τ for 99.99

Figure 3, Calculations for period

As we can see from these calculations, we need approximately 4.6τ for 99.99% of the charge to be dissipated..? (google says 99.3%?). However, due to the nature of logarithm, theoretically the charge will never be zero. As such, we can just pick some multiple of τ to set as the period so that sufficient (99% maybe) dissipation is satisfied; the rate of dissipation gets slower and slower so it will not be worth taking all that extra time for a few thousandths of a percent.

I will say that 5τ is reasonable to set as the period for discharge. NOTE: We need to multiply this by 2 to account for the charging portion; since the period of a square wave includes one portion of both the 'on' and 'off' states of the source. So; we will allow the total period to be 10τ, or for our circuit, 5 seconds. The desired frequency is the inverse of that: $f=\frac{1}{5s}=0.2Hz$.

3. If the source is an AC signal, is the circuit a high pass or low pass filter?

We can determine whether it is a low pass or high pass filter by examining the circuit behaviour at high and low frequencies. This is because for AC circuits, capacitors exhibit a behaviour known as impedence. This value is similar to resistance in DC circuits, and is reliant on the frequency of the AC source:

$$X_c=\frac{1}{2\pi fC}$$

We can take a look at either extrema, which I will look at the case for high frequency first:

As frequencies get super high, the denominator gets super large and so the impedence approaches 0, meaning the capacitor will act as a short circuit to the ground, and (+) will get no power. So $V_in \neq V_out$ (since V_out is measured at (+) in figure 1). As such, the original signal does not pass through, meaning high frequency sources are filtered out.

As frequencies get super low, the denominator gets super small and so the impedence approaches a very large number, meaning the capacitor may start to act as an open circuit (for sufficiently small f). Because of this, the signal does not go directly to (-), rather it passes to (+) as also seen in figure 1; The full original source/signal passes through the system. So, low frequency sources are let through!

Given these two results, we can conclude that this circuit functions as a low-pass filter (low frequency is passed). we can also conclude that if the circuit was reversed, so that the resistor came after the capacitor, it would behave as a high pass filter instead.

What about the f_{3dB} value? We know from the tutorial that this value is when the output power in a circuit drops to $\frac{1}{\sqrt{2}}$ of its maximum value. This is the calculation for this value (C is 500nF, R is 1 $M\Omega$):

$$f_{3dB}=\frac{1}{2\pi RC}$$
$$=\frac{1}{2\pi 0.5}$$
$$=\frac{1}{\pi}Hz$$

This value is quite clean!

Note to self: tag troubleshooting with TS

#2 Experiment

##2.1 measuring unknown R and C

The first measurement I will complete is measuring the unknown resistor value. This can be done by making a voltage divider with a known resistor. First I will set up the circuit with a 5V DC power source, and have the two resistors in series. From this, I can measure the voltage drop accross the first known resistor to determine the current in the circuit, and then use that with the voltage drop across the unknown resistor to calculate the value of the unknown resistor.

First, I measured the value of the known resistor just to make sure of its value. The DMM recorded $0.989\pm\frac{0.001}{\sqrt{3}}\ K\Omega$.

Then, I placed it back into the circuit in series with the unknown resistor and started a 5V power source in the waveforms software with the AD3. I have included a diagram of my scope settings, and the circuit + diagram below:

```
In [21]: InlineImage('unknowncircut.png', width=12, height=8, caption= "Figure 4, circuit with known and unknown resistor in series. 5V power source.", fs=12);
         InlineImage('name1.png', width=12, height=8, caption= "Figure 5, Diagram of circuit setup with known and unknown resistor in series", fs=12);
         InlineImage('5volted.png', width=12, height=8, caption= "Figure 6, Scope power source settings", fs=12);
```

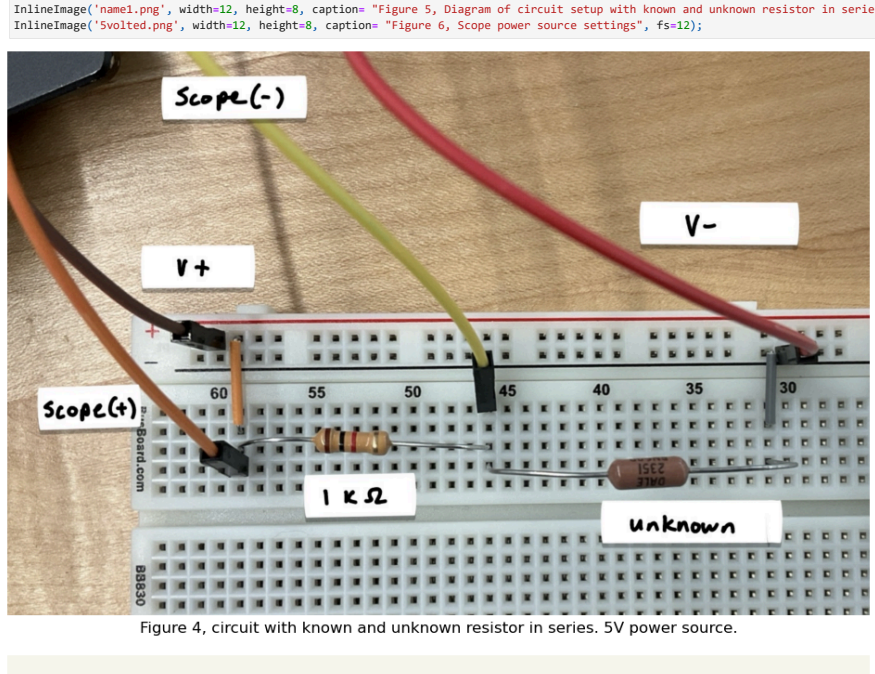

Scope(-)
V-
V+
Scope(+)
1 KΩ
unknown

Figure 4, circuit with known and unknown resistor in series. 5V power source.

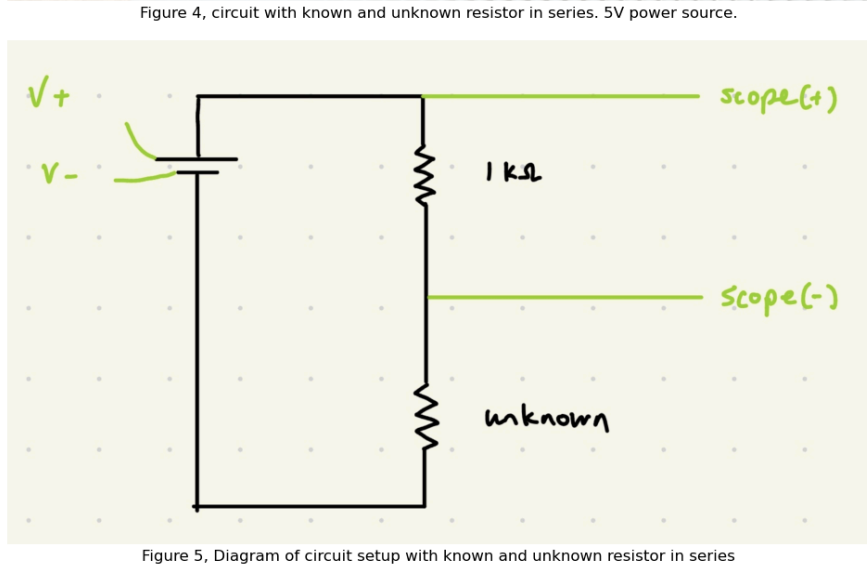

V+
V-
Scope(+)
1 KΩ
Scope(-)
unknown

Figure 5, Diagram of circuit setup with known and unknown resistor in series

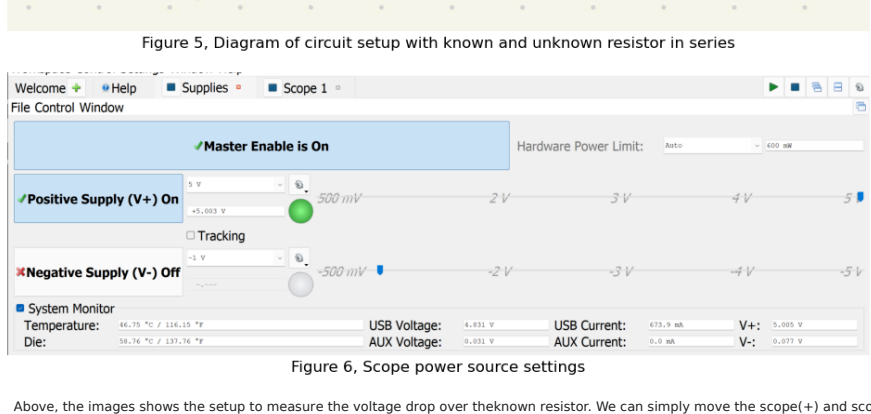

Welcome
Help
Supplies
Scope 1
File Control Window
Master Enable is On
Hardware Power Limit:
Positive Supply (V+) On
Tracking
Negative Supply (V-) Off
System Monitor
Temperature:
Die:
USB Voltage:
AUX Voltage:
USB Current:
AUX Current:
V+:
V-:

Figure 6, Scope power source settings

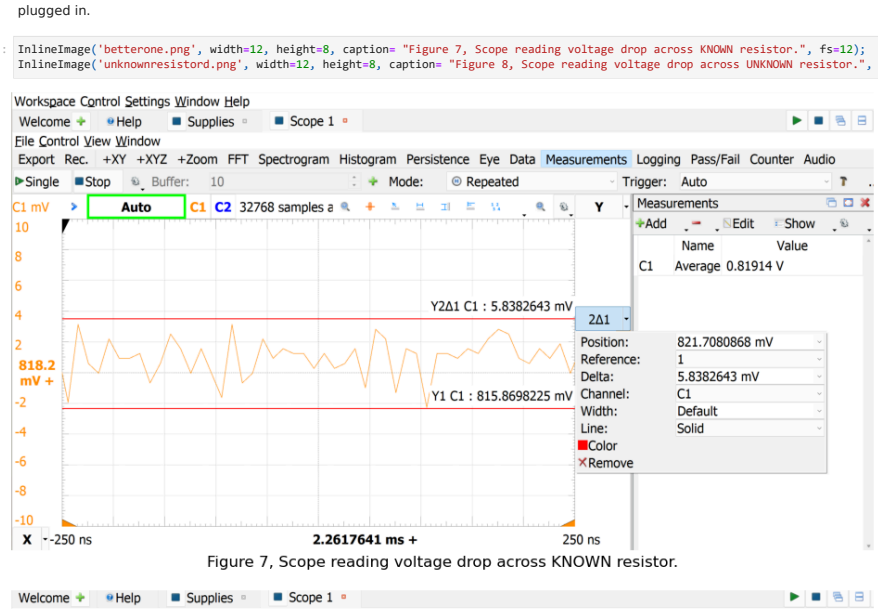
Above, the images shows the setup to measure the voltage drop over theknown resistor. We can simply move the scope(+) and scope(-) to be across the unknown resistor instead.

Here, as we can see, the scope measured the voltage accross the known resistor to be 0.81814 volts. This is very precise, and there should be an uncertainty. As can be seen in figure 7 below, we can see the range of fluctuation. Below this image I will include my calculations for the uncertainty of the value for voltage drop.

TS: to see the waveform flucuation, I had to make the scale division smaller and change the offset. I ended up having an offset of approx. -0.818 V, and a scale of 2mV/div, and then I was able to see the fluctuation.

TS: I verified this voltage drop with the DMM, but I was getting wonky readings. At first, I realized it was on the AC mode for voltage and I changed it to DC, but it was still off. Then, I realized the measuring probes were just not properly plugged in.

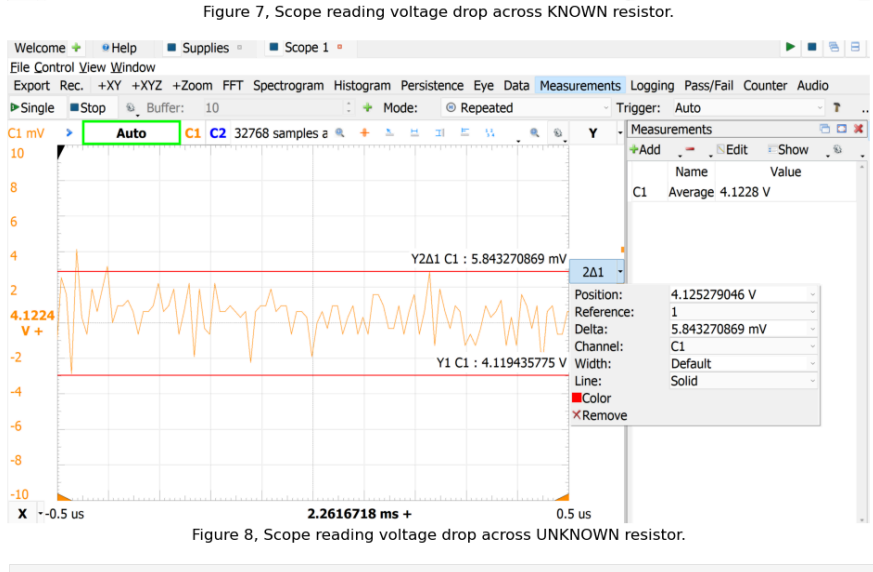
```
In [28]: InlineImage('betterone.png', width=12, height=8, caption= "Figure 7, Scope reading voltage drop across KNOWN resistor.", fs=12);
         InlineImage('unknownresistord.png', width=12, height=8, caption= "Figure 8, Scope reading voltage drop across UNKNOWN resistor.", fs=12);
```


Workspace Control Settings Window Help
Export Rec. +XY +XYZ +Zoom FFT Spectrogram Histogram Persistence Eye Data Measurements Logging Pass/Fail Counter Audio
Single Stop Buffer: 10 Mode: Repeated Trigger: Auto
C1 mV
Auto
C1 C2 32768 samples
Y2Δ1 C1 : 5.8382643 mV
Y1 C1 : 815.8698225 mV
X −250 ns
2.2617641 ms +
250 ns

Measurements: C1 Average 0.81914 V

Position: 821.7080868 mV
Reference: 1
Delta: 5.8382643 mV
Channel: C1
Width: Default
Line: Solid

Figure 7, Scope reading voltage drop across KNOWN resistor.


Export Rec. +XY +XYZ +Zoom FFT Spectrogram Histogram Persistence Eye Data Measurements Logging Pass/Fail Counter Audio
Single Stop Buffer: 10 Mode: Repeated Trigger: Auto
C1 mV
Auto
C1 C2 32768 samples
Y2Δ1 C1 : 5.843270869 mV
Y1 C1 : 4.119435775 V
X −0.5 us
2.2616718 ms +
0.5 us

Measurements: C1 Average 4.1228 V

Position: 4.125279046 V
Reference: 1
Delta: 5.843270869 mV
Channel: C1
Width: Default
Line: Solid

Figure 8, Scope reading voltage drop across UNKNOWN resistor.

```
In [30]: InlineImage('uncertainns.png', width=12, height=8, caption= "Figure 8, uncertainty from scope reading calculations for known and unknown resistors", fs=12);
```

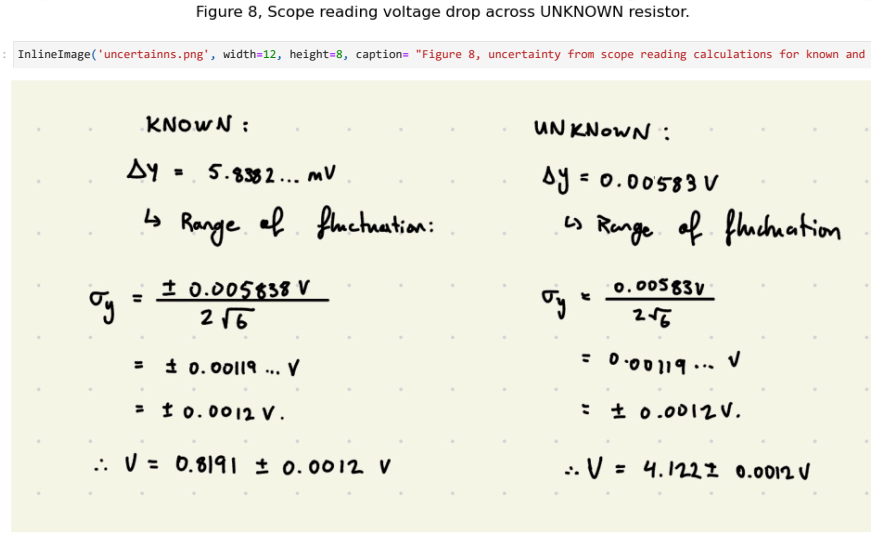

KNOWN :
ΔY = 5.8382... mV
↳ Range of fluctuation:
σy = ± 0.005838 V / 2√6
= ± 0.00119 ... V
= ± 0.0012 V.
∴ V = 0.8191 ± 0.0012 V
UNKNOWN :
Δy = 0.00583 V
↳ Range of fluctuation
σy = 0.00583V / 2√6
= 0.00119 ... V
= ± 0.0012V.
∴ V = 4.1222 0.0012 V

Figure 8, uncertainty from scope reading calculations for known and unknown resistors

TS This is quite a precise value, maybe a bit too precise. I will check the uncertainty of the volt meter, and take whichever is higher: the volt meter gives a reading of 0.814V, with no fluctuation. The uncertainty is hence: $\frac{0.001}{\sqrt{3}}=\pm0.000057...V$. This value seems even more precise, so I will go with the oscilloscope uncertainty to propogate forward.

NOTE: The voltage source on my AD3 also has an uncertainty as it fluctuates. The fluctuation is by 0.008V (between 5.003 and 5.011), so I will divide this value by $\sqrt{3}$ (digital reading) to find the uncertainty for the source: Uncertainty of voltage is $\pm$ 0.0046...V.

Now, to actually find the value of the unknown resistor:

We have a circuit in series with each other, so we know the current accross both resistors is the same. I will start by using KVL to make an expression for the circuit, then I will write current in terms of the voltage over the unknown resistor and then solve for $R_{unknown}$! Below are my calculations. I have also propogated uncertainty.

```
In [32]: InlineImage('vunknown.png', width=12, height=8, caption= "Figure 9, Calculation for unknown resistance value", fs=12);
         InlineImage('uncertaid.png', width=12, height=8, caption= "Figure 10, Uncertainty propogation for unknown resistance value", fs=12);
```

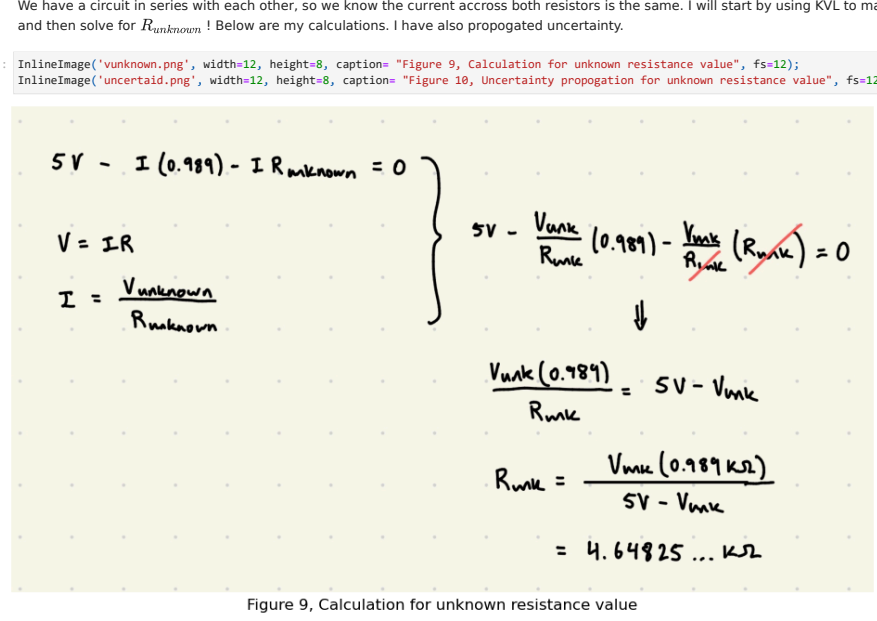

5V − I(0.989) − I Runknown = 0
V = IR
I = Vunknown / Runknown
5V − (Vunk/Runk)(0.989) − (Vunk/Runk)(Runk) = 0
⇓
Vunk(0.989) / Runk = 5V − Vunk
Runk = Vunk(0.989 kΩ) / (5V − Vunk)
= 4.64825 ... kΩ

Figure 9, Calculation for unknown resistance value

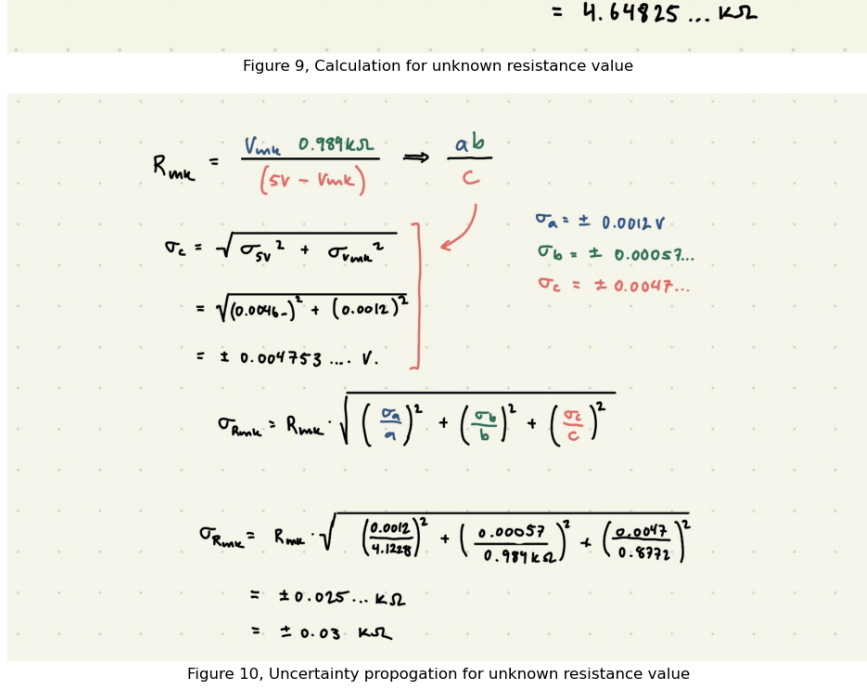

Runk = Vunk 0.989kΩ / (5V − Vunk) ⟹ ab/c
σc = √(σ5V² + σVunk²)
= √((0.0046..)² + (0.0012)²)
= ± 0.004753 .... V.
σa = ± 0.0012 V
σb = ± 0.000057...
σc = ± 0.0047...
σRunk = Runk · √((σa/a)² + (σb/b)² + (σc/c)²)
σRunk = Runk · √((0.0012/4.1228)² + (0.000057/0.989kΩ)² + (0.0047/0.8772)²)
= ±0.025 ... kΩ
= ± 0.03 kΩ

Figure 10, Uncertainty propogation for unknown resistance value

As such, I the final value for the unkown resistor is $4.65\pm0.03k\Omega$.

Now, for the second part: measuring unknown C:

NOTE: to find the frequency, we can repeat the prelab calculations. However, we will be using a $1k\Omega$ resistor this time rather than a $1M\Omega$ so we should change the values. Nevermind. on second thought, we do not know the value of capacitance, so we cannot actually calculate the required frequency. So, I will mess around with the settings in the AD3 (both the source wave frequency and the scope window settings) until I see a good waveform.

I will do this by generating a square wave with a circuit that has the unknown capacitor with the known resistor value. Then, I will plot the data points of the curve for capacitor discharge, and linearize it in MATLAB (or excel).

With the capacitor discharge formula, I will be able to determine an equation that has tau as part of the slope, and knowing R, I will be able to calculate C. I will do this through last year's phys 159 MATLAB livescript for linearization for ease of plotting and uncertainty propogation. I am inclined towards this method as from previous experimentation in phys159, this method reduced uncertainty and provided better results from more data points as it removes the need to 'guess'/'estimate' the point of ~63% energy remaining from the discharge waveform - roughly 1 τ (which is prone to human error). However, If I am not able to get the slope method to work, I will just estimate the time for the ~63% energy point.

Below, I have my work in finding what I need to plot to acquire the linearized capacitor relationship:

```
In [33]: InlineImage('linear.png', width=12, height=8, caption= "Figure 11, Calculation for linearized capacitance relationship to plot", fs=12);
```

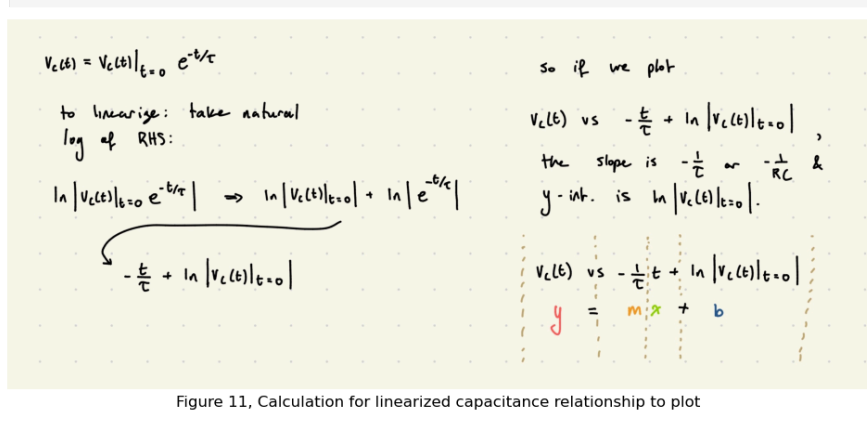

Vc(t) = Vc(t)|t=0 e^(−t/τ)
to linearize: take natural log of RHS:
ln|Vc(t)|t=0 e^(−t/τ)| ⟹ ln|Vc(t)|t=0| + ln|e^(−t/τ)|
−t/τ + ln|Vc(t)|t=0|
so if we plot
Vc(t) vs −t/τ + ln|Vc(t)|t=0| ,
the slope is −1/τ or −1/RC &
y-int. is ln|Vc(t)|t=0|.
Vc(t) vs −(1/τ)t + ln|Vc(t)|t=0|
y = m x + b

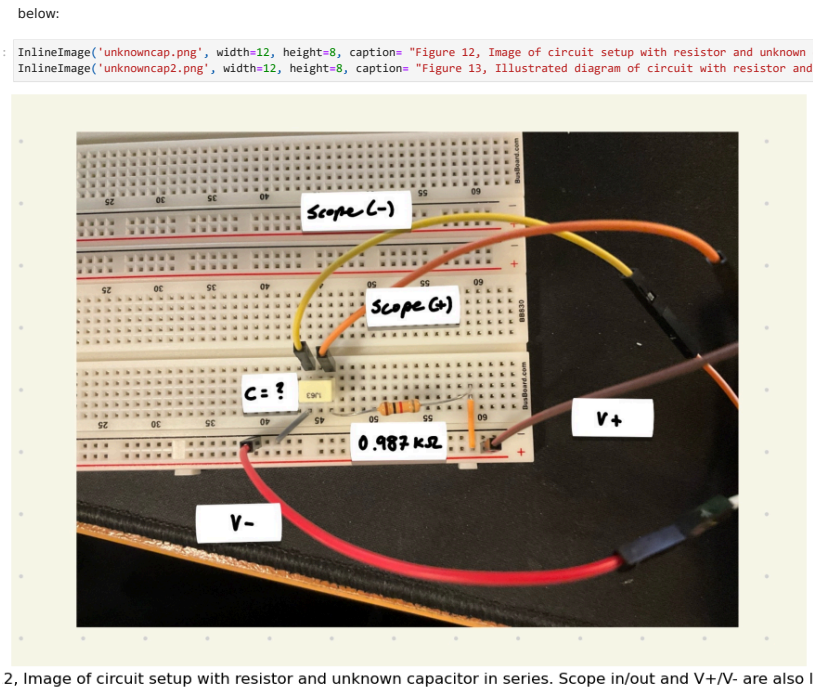
Figure 11, Calculation for linearized capacitance relationship to plot

As we can see, I can simply take the natural log of the right hand side of the equation, which corresponds to the voltage. Then, we can plot this value on the y axis with time on the x axis, and we should get a graph with a negative slope, which will have the value of $-\frac{1}{RC}$. Then, since I will use a resistor with a known resistance ($1k\Omega$), I can simply rearrange and calculate for C:

$$C=\frac{\frac{1}{R}}{slope}=\frac{\frac{1}{R}}{\frac{1}{RC}}$$

Now, I will actually build the circuit to do this. I will have 1 capacitor (unknown value) and a resistor in series, with an AC square wave power source from the AD3. I have included images of the ciruit along with an illustrated diagram below:

```
In [34]: InlineImage('unknowncap.png', width=12, height=8, caption= "Figure 12, Image of circuit setup with resistor and unknown capacitor in series. Scope in/out and V+/V- are also labelled.", fs=12);
         InlineImage('unknowncap2.png', width=12, height=8, caption= "Figure 13, Illustrated diagram of circuit with resistor and unknown capacitor in series.", fs=12);
```


Scope(-)
Scope(+)
C=?
0.987 kΩ
V+
V-

2, Image of circuit setup with resistor and unknown capacitor in series. Scope in/out and V+/V- are also l

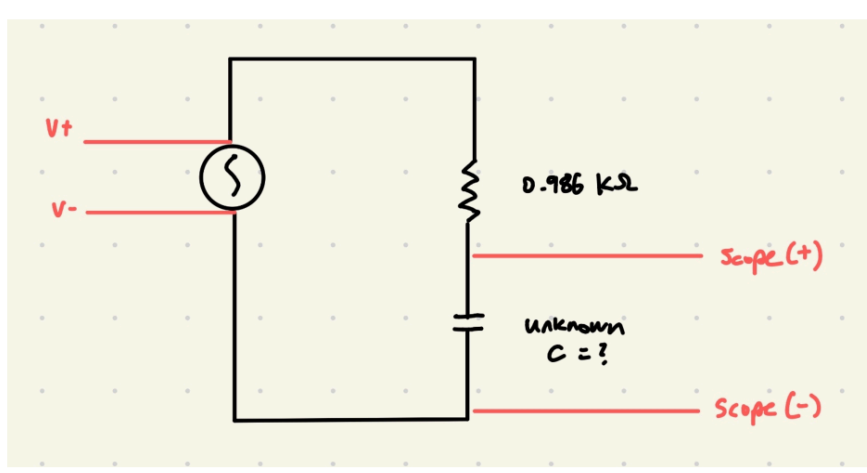

V+
V-
0.986 kΩ
Scope(+)
unknown C=?
Scope(-)

Figure 13, Illustrated diagram of circuit with resistor and unknown capacitor in series.

With this done, I will now begin experimenting with scope settings to find a good view of the waveform. The power source is a 1 Volt amplitude AC square wave. I have set the offset to 1V, so that the AC source ranges from 0 to 2 volts, simulating a switch for a DC source. This will allow the capacitor to charge or discharge in the time the switch is on/off, and I can then observe the discharge waveform and graph it.

TS Initially, the frequency is set to 1kHz. However, the waveform appeared as pretty much a straight line with static. I think this is because the ac source is too fast for the capacitor to charge, so I will decrease the frequency. The I tried a range of frequencies, going down in steps from 1 kHz to 1 mHz. This doesn't seem to work. I am confused now, but I think it is not the speed of the source now. Rather, I will investigate the scope window settings.

TS First, I changed the scale of the scope to view the waveform closer up. I got the waveform centered at 1 mV / div. However, it still looks like a straight line with static. However, in confusion and staring at the screen blankly, I by chance noticed the waveform moving down further and further on the screen. Does this mean that the capacitor is gradually discharging?? but at a super super slow rate??? Maybe I need to increase the voltage of the power source. With a constant R value, if V goes up, I also goes up. more current will help it charge faster I think.

TS Okay, so changing the voltage amplitude to 2.5Volts did not work either. I see no visible change in the waveform.

TS Now I am doubting my connections. I will double check if the circuit is closed.

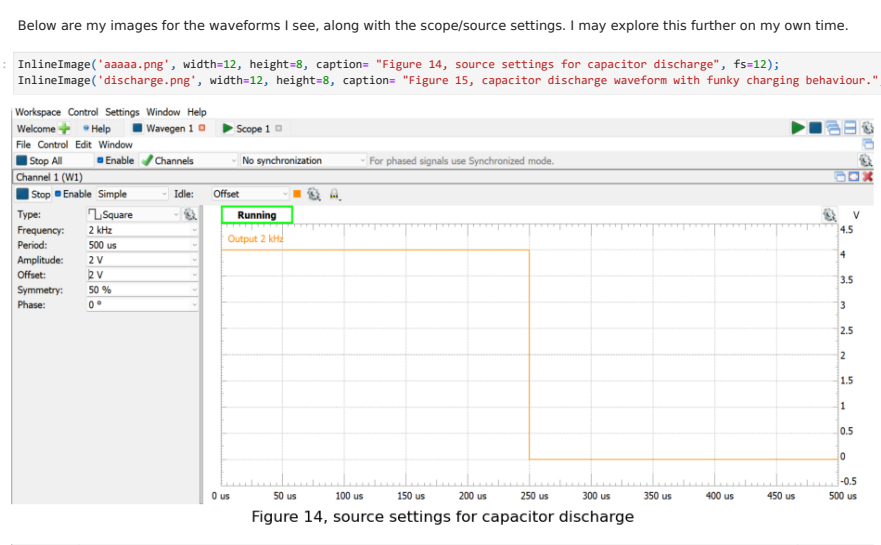
TS halleluja!!! Okay big discovery. I was re-plugging in the V+ terminal and I happened to have my waveform scope still running (usually I pause the power supply in caution), and I see that my earlier suspicion was correct; the waveform moving down turned out to be the capacitor slowly discharging. I can disconnect the V+ to see the full discharge, although it is quite slow; it looks like a straight line. I can also re-connect it to briefly see the charging of the capacitor. This seems super strange; why is the charging so much faster than the discharge?

TS maybe I need to turn the voltage down and not up? Ok this seems to change nothing.

TS I went for a break and came back, and adjusted the settings again. This time, I was able to see the discharge curve as I would expect it. However, the charging is still quite a lot faster, and I am still unsure as to why. However, since I only need the discharging curve for the capacitor, I can move forward with these waveforms and data.

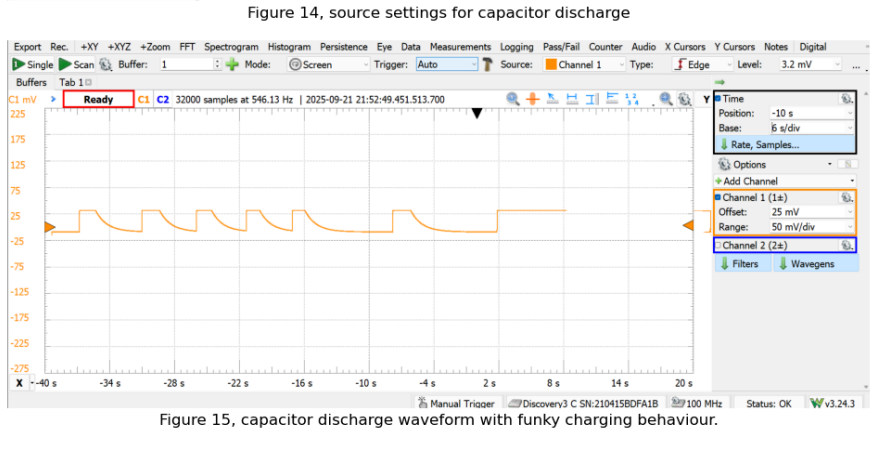
Below are my images for the waveforms I see, along with the scope/source settings. I may explore this further on my own time.

```
In [35]: InlineImage('aaaaa.png', width=12, height=8, caption= "Figure 14, source settings for capacitor discharge", fs=12);
         InlineImage('discharge.png', width=12, height=8, caption= "Figure 15, capacitor discharge waveform with funky charging behaviour.", fs=12);
```


Workspace Control Settings Window Help
Welcome
Help
Wavegen 1
Scope 1
Stop All
Enable
Channels
No synchronization
Channel 1 (W1)
Running
Output 2 kHz
0 us
500 us

Type: Square
Frequency: 2 kHz
Period: 500 us
Amplitude: 2 V
Offset: 2 V
Symmetry: 50 %
Phase: 0 °

Figure 14, source settings for capacitor discharge


Export Rec. +XY +XYZ +Zoom FFT Spectrogram Histogram Persistence Eye Data Measurements Logging Pass/Fail Counter Audio X Cursors Y Cursors Notes Digital
Single Stop Buffer: 1 Mode: Screen Trigger: Auto Source: Channel 1 Type: Edge Level: 3.2 mV
Ready
C1 C2 32000 samples at 546.13 Hz | 2025-09-21 21:52:49.451.513.700
X −40 s
20 s
Manual Trigger
Discovery3 C SN:210415BDFA1B
100 MHz
Status: OK
v3.24.3

Time Position: -10 s
Base: 6 s/div
Channel 1 (1x) Offset: 25 mV
Range: 50 mV/div

Figure 15, capacitor discharge waveform with funky charging behaviour.

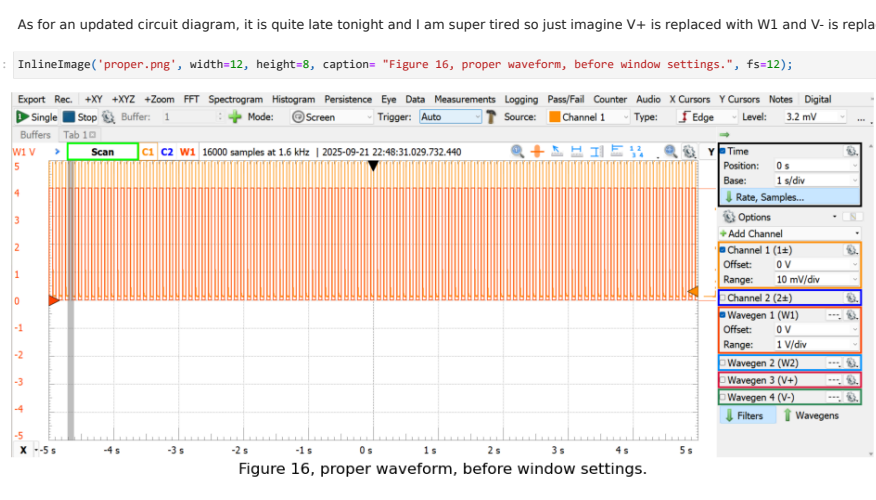
TS wow, I finally figured out my problem. I was wrongly using the V+ and V- as the source rather than W1 and gnd. This error can probably be chalked up to reading comprehension..? I toggled this menu for the scope to see different things, such as the power, and I found out that for waves and not DC sources, I should be using W1 and gnd. This actually makes a lot more sense now and I can see the proper charge/discharges!!!!!

Ok for now, I will mess with the settings again just to get a good image of the waveform, and come back and paste them in. This took me a while to figure out, I feel quite triumphunt but also frustrated at how simple this error was. Needless to say I will not be making this mistake in the future.

Anyways, for now, here is an image of how using W1 and GND looks. Its obviously much better.

As for an updated circuit diagram, it is quite late tonight and I am super tired so just imagine V+ is replaced with W1 and V- is replaced with GND in my original diagram.

```
In [36]: InlineImage('proper.png', width=12, height=8, caption= "Figure 16, proper waveform, before window settings.", fs=12);
```

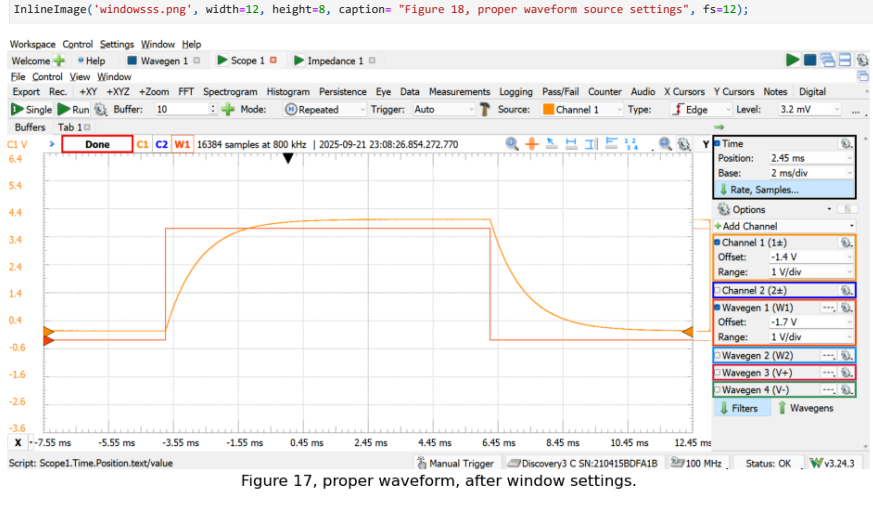

Export Rec. +XY +XYZ +Zoom FFT Spectrogram Histogram Persistence Eye Data Measurements Logging Pass/Fail Counter Audio X Cursors Y Cursors Notes Digital
Single Stop Buffer: 1 Mode: Screen Trigger: Auto Source: Channel 1 Type: Edge Level: 3.2 mV
Scan
C1 C2 W1 16000 samples at 1.6 kHz | 2025-09-21 22:48:31.029.732.440
X −5 s
5 s

Time Position: 0 s
Base: 1 s/div
Channel 1 (1x) Offset: 0 V
Range: 10 mV/div
Wavegen 1 (W1) Offset: 0 V
Range: 1 V/div

Figure 16, proper waveform, before window settings.

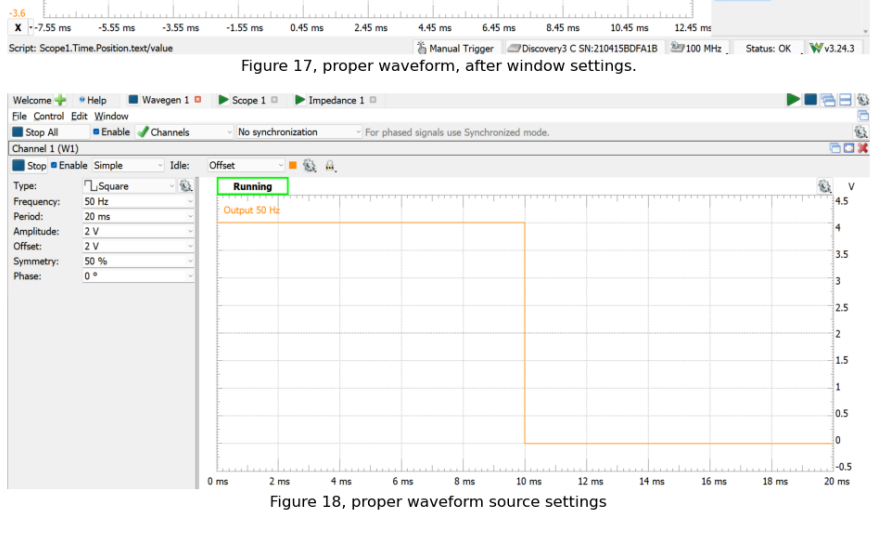
I have configured the windows nicely now; below are my waveforms :D

```
In [38]: InlineImage('goodest.png', width=12, height=8, caption= "Figure 17, proper waveform, after window settings.", fs=12);
         InlineImage('windowsss.png', width=12, height=8, caption= "Figure 18, proper waveform source settings", fs=12);
```


Workspace Control Settings Window Help
Welcome
Help
Wavegen 1
Scope 1
Impedance 1
Single Run Buffer: 10 Mode: Repeated Trigger: Auto Source: Channel 1 Type: Edge Level: 3.2 mV
Done
C1 C2 W1 16384 samples at 800 kHz | 2025-09-21 23:08:26.854.272.770
X −7.55 ms
12.45 ms
Script: Scope1.Time.Position.text/value
Manual Trigger
Discovery3 C SN:210415BDFA1B
100 MHz
Status: OK
v3.24.3

Time Position: 2.45 ms
Base: 2 ms/div
Channel 1 (1x) Offset: -1.4 V
Range: 1 V/div
Wavegen 1 (W1) Offset: -1.7 V
Range: 1 V/div

Figure 17, proper waveform, after window settings.


Welcome
Help
Wavegen 1
Scope 1
Impedance 1
Stop All
Enable
Channels
No synchronization
Channel 1 (W1)
Running
Output 50 Hz
0 ms
20 ms

Type: Square
Frequency: 50 Hz
Period: 20 ms
Amplitude: 2 V
Offset: 2 V
Symmetry: 50 %
Phase: 0 °

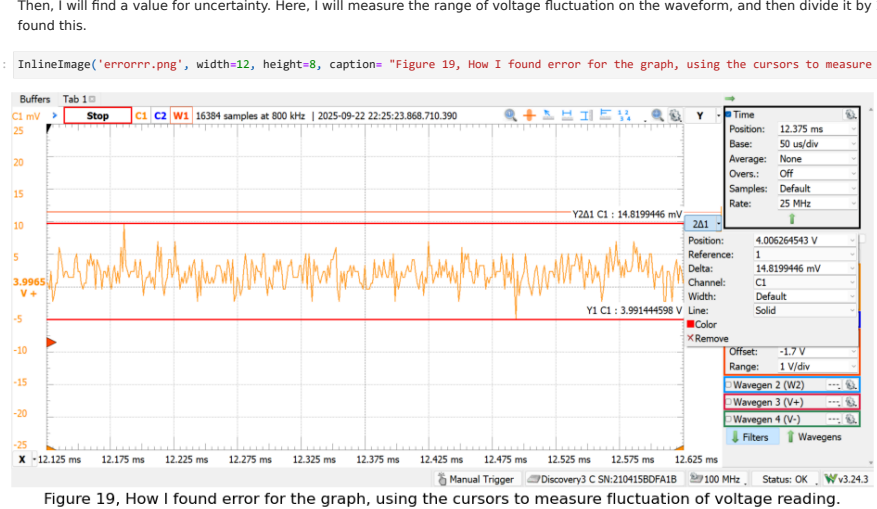
Figure 18, proper waveform source settings

Now that we have the waveforms, we can move on to plotting the data. I have used the matlab livescript from last year's 159 course, and selected a range so I only have the discharge curve (minimal flat points as possible). I will then linearize this, and from the relationship in figure 11 obtain the value for C. The reason I am doing this method as opposed to the one where I measure where approx. 63% of energy has dissipated for 1 time constant is because from last year, it was proven to have less error and be more accurate due to the high volume of data points.

The first step in this process is importing the data to matlab. **marked as done**

Then, I will find a value for uncertainty. Here, I will measure the range of voltage fluctuation on the waveform, and then divide it by $2\sqrt{3}$ since it is a digital measurement. This gives me a value of $\pm4.28mV$. See figure 19 below for how I found this.

```
In [39]: InlineImage('errorrr.png', width=12, height=8, caption= "Figure 19, How I found error for the graph, using the cursors to measure fluctuation of voltage reading.", fs=12);
```


Buffers Tab 1
Stop
C1 C2 W1 16384 samples at 800 kHz | 2025-09-22 22:25:23.868.710.390
Y2Δ1 C1 : 14.8159446 mV
Y1 C1 : 3.991444598 V
X −12.125 ms
12.625 ms
Manual Trigger
Discovery3 C SN:210415BDFA1B
100 MHz
Status: OK
v3.24.3

Time Position: 12.375 ms
Base: 50 us/div
Average: None
Overs.: Off
Samples: Default
Rate: 25 MHz

Position: 4.006264543 V
Reference: 1
Delta: 14.8159446 mV
Channel: C1
Width: Default
Line: Solid

Figure 19, How I found error for the graph, using the cursors to measure fluctuation of voltage reading.

Now I can graph. Here is my graph without linearization:

```
In [44]: InlineImage('matlab.png', width=16, height=8, caption= "Figure 20, non linearized capacitor waveform. Section to be graphed is highlighted in red. \n Error bars present.", fs=12);
```

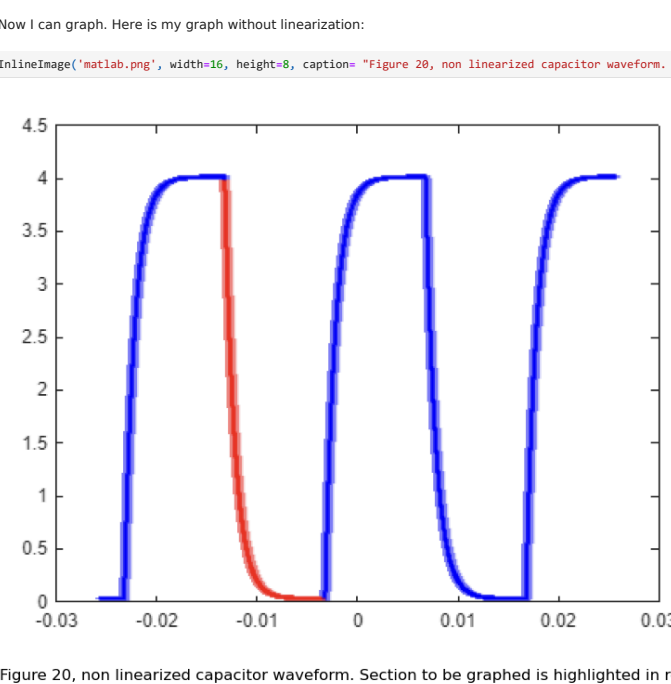


Figure 20, non linearized capacitor waveform. Section to be graphed is highlighted in red.
Error bars present.

TS Oops, looks like I got a bit too much of the flat part. as we can see below: the graph is mostly linear, but changes at the end. I will remedy this by selecting a smaller range of values and have less of the flat section.

Here is what my graph looked like before with original range:

```
In [45]: InlineImage('curvey.png', width=16, height=8, caption= "Figure 21, slightly curved graph of linearized data.", fs=12);
```

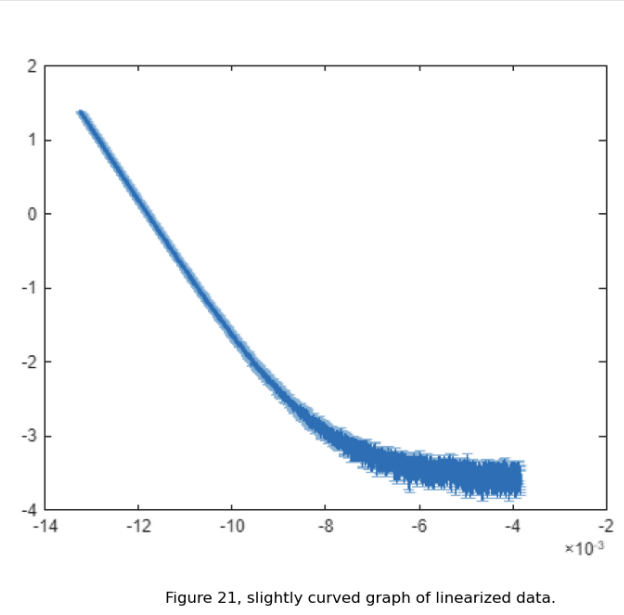


Figure 21, slightly curved graph of linearized data.

Here is what a better range looks like: (I used roughly 1200 data points for this one, compared to the 2000 I initially had with the curved one)

```
In [46]: InlineImage('lineard.png', width=16, height=8, caption= "Figure 22, successfully linearezed graph!", fs=12);
```

```
errorbar(time2,logV2,del_logV2,'.')
```

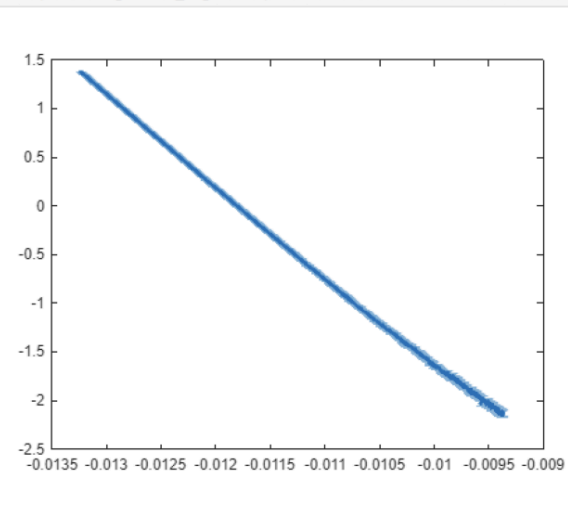


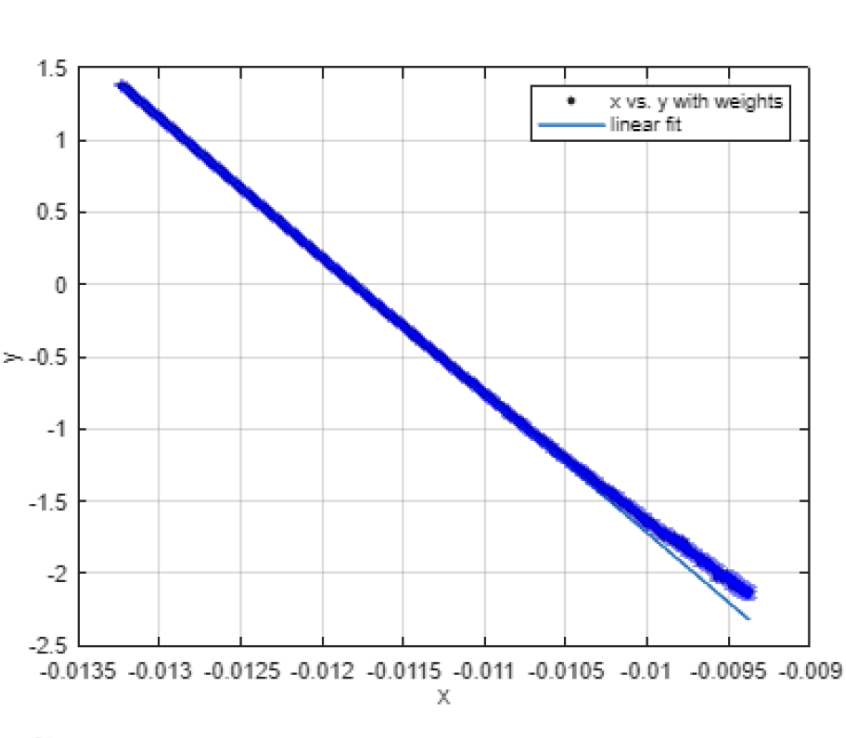
Figure 22, successfully linearezed graph!

Note: X axis is in the negatives because of my offset in oscilloscope. However, in this situation time is relative so the values still work.

Below is my linear fit for this graph. This was also obtained with the matlab livescript.

```
In [50]: InlineImage('linear_fit.png', width=16, height=16, caption= "Figure 23, linearized graph with linear fit and slope values", fs=12);
```

```
[fitresult, gof]=Phys159_linfit(time2,logV2,del_logV2)
```



```
fitresult =
     General model:
     fitresult(x) = m*x+b
     Coefficients (with 95% confidence bounds):
       b =      -11.28  (-11.29, -11.27)
       m =      -956.3  (-957.2, -955.4)
gof = struct with fields:
           sse: 9.1059e+03
       rsquare: 0.9997
           dfe: 1234
    adjrsquare: 0.9997
          rmse: 2.7165
```

Figure 23, linearized graph with linear fit and slope values

From the software, I got that the uncertainty for the slope is $\pm0.1601s^{-1}$. Here, I will calculate the propogated uncertainty value as well as the value of C:

```
In [51]: InlineImage('calculationn.png', width=16, height=16, caption= "Figure 24, Calculation for the value of C, with uncertainty calculation.", fs=12);
```

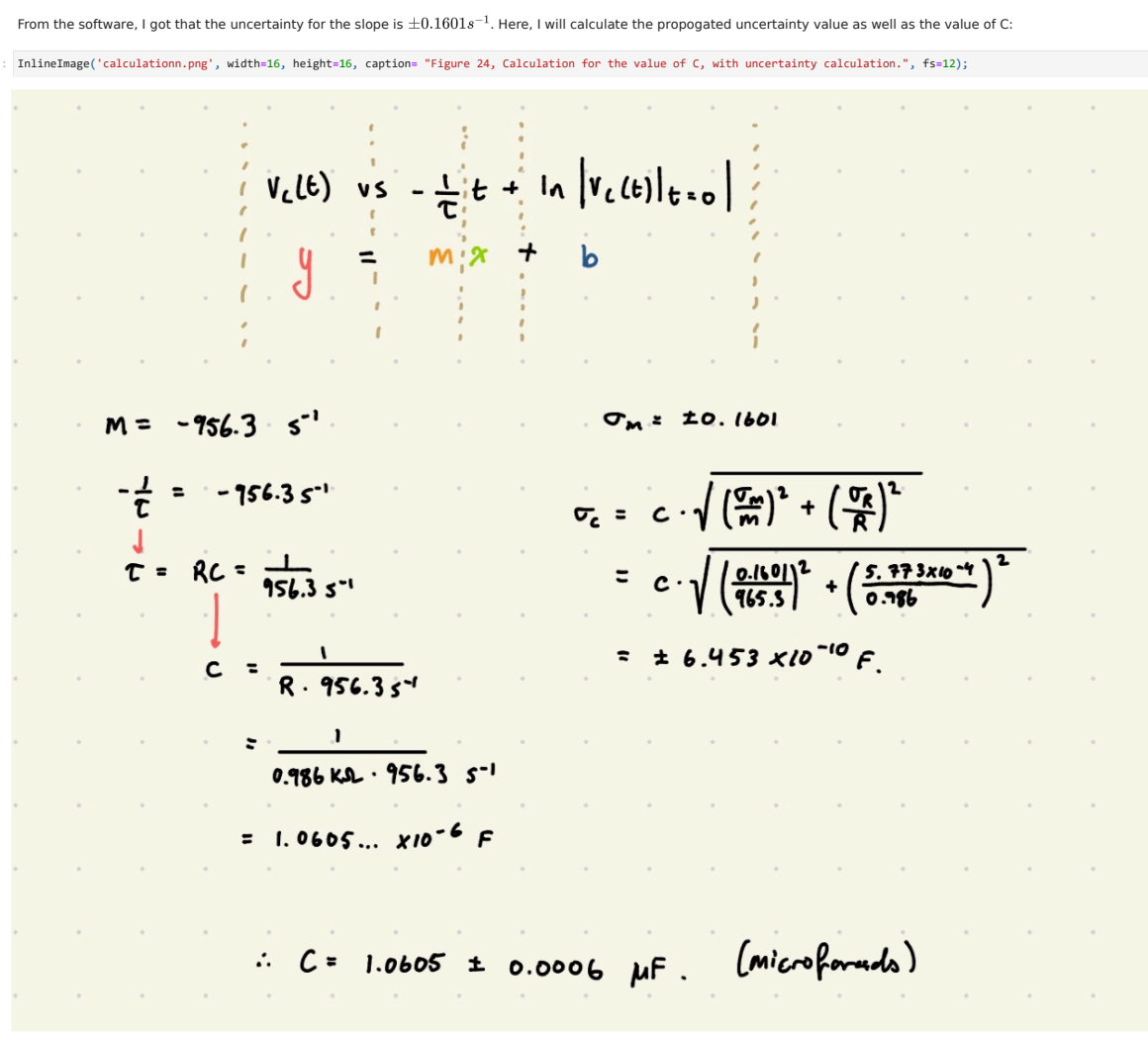

Vc(t) vs −(1/τ)t + ln|Vc(t)|t=0|
y = m x + b
M = −956.3 s⁻¹
−1/τ = −956.3 s⁻¹
τ = RC = 1 / 956.3 s⁻¹
C = 1 / (R · 956.3 s⁻¹)
= 1 / (0.986 kΩ · 956.3 s⁻¹)
= 1.0605... ×10⁻⁶ F
σm = ±0.1601
σc = c · √((σm/m)² + (σR/R)²)
= c · √((0.1601/956.3)² + (5.773×10⁻⁴/0.986)²)
= ± 6.453 ×10⁻¹⁰ F.
∴ C = 1.0605 ± 0.0006 μF. (microfarads)

Figure 24, Calculation for the value of C, with uncertainty calculation.

As we can see, the reading comes out to be around 1 microfarad, which seems reasonable for a typical resistor. it is also a clean value, sitting very close to 1 μF. The uncertainty is also quite precise, compared to the value I got from the fluctuations. The propogation of uncertainty into a larger data set means that with each added value, it causes a gradual convergence towards the actual true value of what I am trying to measure. Hence, with over a thousand data points and each one serving to minimize error, I was able to obtain quite a precise result. However, I will note that this result is not from a direct measurement, rather calculations stemming from values retrieved from measurements. This means that there is possibility for error within the measurements themselves and hence the measured values, so of course, it will not be as accurate as something to measure the value even if it is super precise.

##2.2 measuring the 3dB cutoff for a simple low pass filter

The objective for this section of the lab is to construct a low pass filter with a 500nF capacitor and 100 $k\Omega$ resistor in series. I will then use a sinusoidal source and vary its frequency, to measure its 3dB frequency. Recall the prelab, where we said that the 3dB frequency is the frequency at which the output power for a circuit is $\frac{1}{\sqrt{2}}$ of its maximum value. For a circuit, it is given by:

$$f_{3dB}=\frac{1}{2\pi RC}$$

For our circuit:

$$f_{3dB}=\frac{1}{2\pi(100k\Omega)(500nF)}$$
$$=3.183...hz$$

for a 5V source, this should occur such that the output power is approximately 3.53555... V. (divide 5V by $\sqrt{2}$)

Now, to construct a low pass filter, we have 1 μF capacitors. We can use two in series, which will be shown in my circuit diagram and calculations in the following diagram. For a low pass filter, we should also place the resistor ahead of the capacitor(s). This is because for low frequency, the capacitor's impedence is super high, so the capacitor lets little to no current through. As such, there is no short circuit to the ground (-), and most/all of the current/power goes to the (+), aka the output.

```
In [54]: InlineImage('diagramed.png', width=12, height=16, caption= "Figure 25, Low pass filter setup with 2x 1uF capacitors \nto emulate a 500nF C. R = 100 kiloOhms..", fs=12);
         InlineImage('dogman.png', width=12, height=16, caption= "Figure 26, Circuit diagram for low pass filter setup.", fs=12);
```

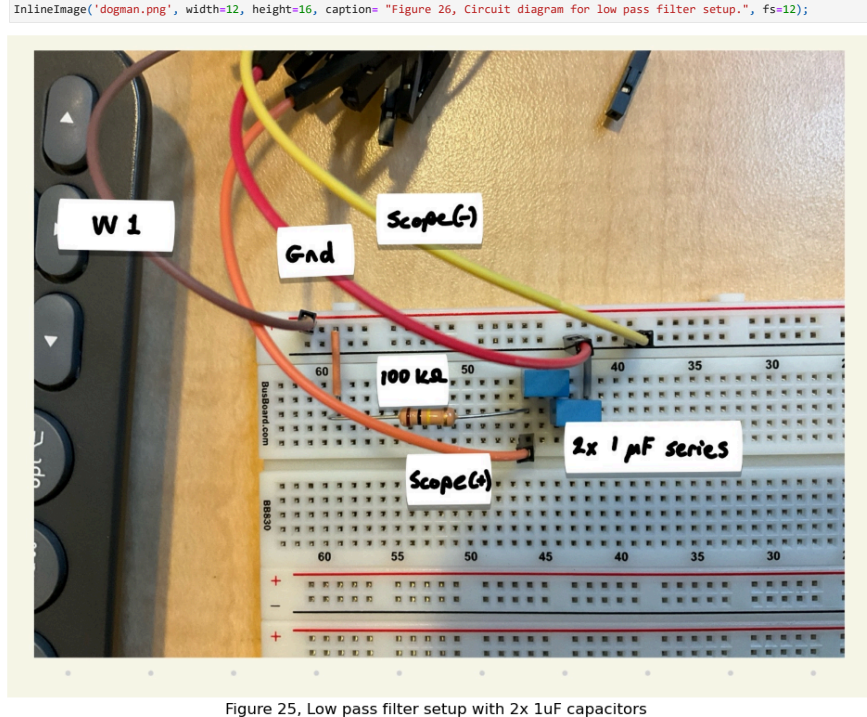

W1
Gnd
Scope(-)
100 kΩ
2x 1 μF series
Scope(+)

Figure 25, Low pass filter setup with 2x 1uF capacitors
to emulate a 500nF C. R = 100 kiloOhms..

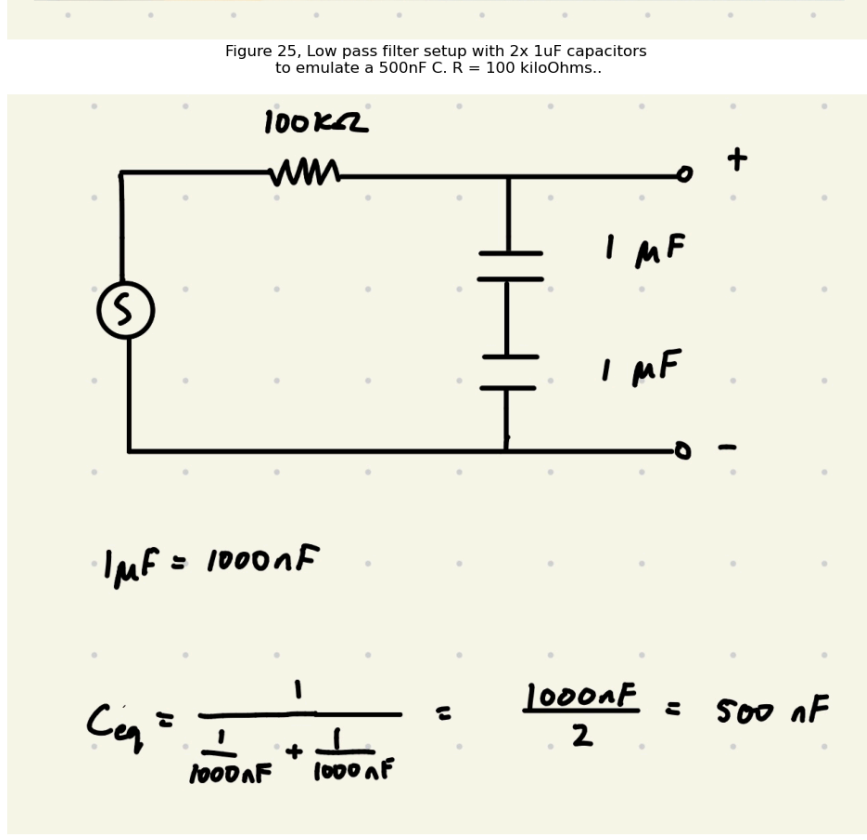
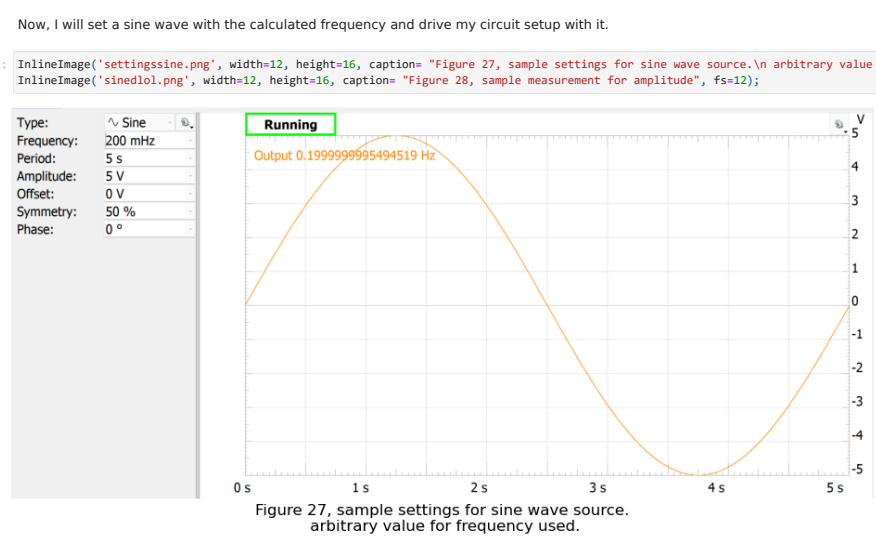

100kΩ
+
1 μF
1 μF
−
1μF = 1000nF
Ceq = 1 / (1/1000nF + 1/1000nF) = 1000nF/2 = 500 nF

Figure 26, Circuit diagram for low pass filter setup.

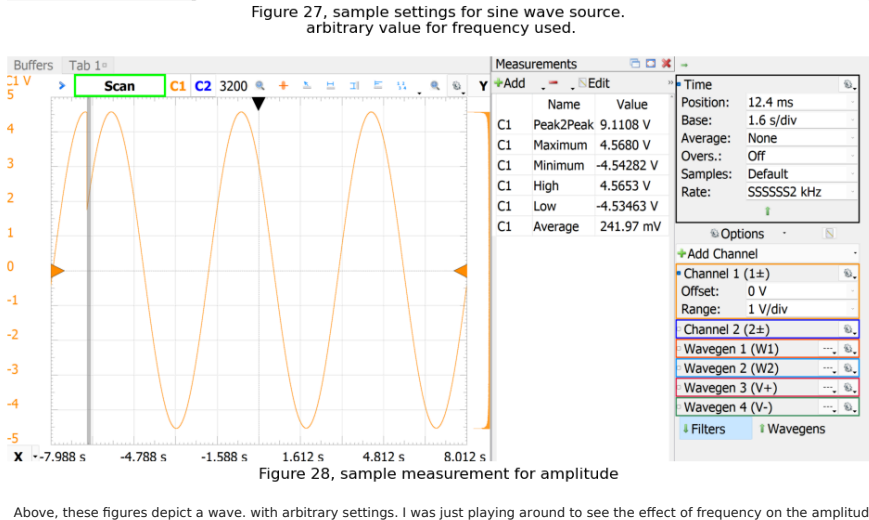
Now, I will set a sine wave with the calculated frequency and drive my circuit setup with it.

```
In [57]: InlineImage('settingssine.png', width=12, height=16, caption= "Figure 27, sample settings for sine wave source.\n arbitrary value for frequency used.", fs=12);
         InlineImage('sinedlol.png', width=12, height=16, caption= "Figure 28, sample measurement for amplitude", fs=12);
```


Running
Output 0.1999999995494519 Hz
0 s
1 s
2 s
3 s
4 s
5 s

Type: Sine
Frequency: 200 mHz
Period: 5 s
Amplitude: 5 V
Offset: 0 V
Symmetry: 50 %
Phase: 0 °

Figure 27, sample settings for sine wave source.
arbitrary value for frequency used.


Buffers Tab 1
Scan
C1 C2 3200
X −7.988 s
-4.788 s
-1.588 s
1.612 s
4.812 s
8.012 s

Name	Value
C1 Peak2Peak	9.1108 V
C1 Maximum	4.5680 V
C1 Minimum	-4.54282 V
C1 High	4.5653 V
C1 Low	-4.53463 V
C1 Average	241.97 mV

Time Position: 12.4 ms
Base: 1.6 s/div
Average: None
Overs.: Off
Samples: Default
Rate: 555552 kHz
Channel 1 (1x) Offset: 0 V
Range: 1 V/div

Figure 28, sample measurement for amplitude

Above, these figures depict a wave. with arbitrary settings. I was just playing around to see the effect of frequency on the amplitude.

As I increase the frequency, the amplitude gets lower and lower. The period also gets lower in accordance to the frequency-period relationship.

I am looking for a voltage amplitude of approx 3.53 volts. I will use my calculated value, and then fine tune using the measurements for maximum and minimum.

TS at a frequency of 3.183 Hz, the voltage amplitude is approx. 3.4 volts. This is a bit lower than expected. should decrease the frequency?

well, not really - I think that my pc is not supplying the full 5V exactly. I will check the tab for the power source info:

```
In [61]: InlineImage('powered.png', width=18, height=16, caption= "Figure 29, voltage output of my computer through usb", fs=12);
```

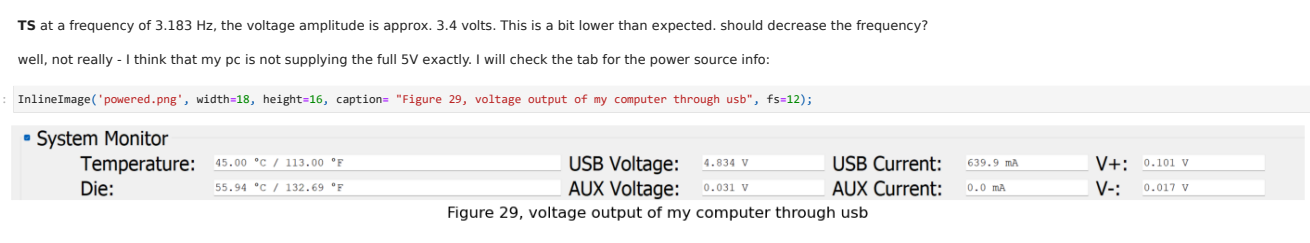
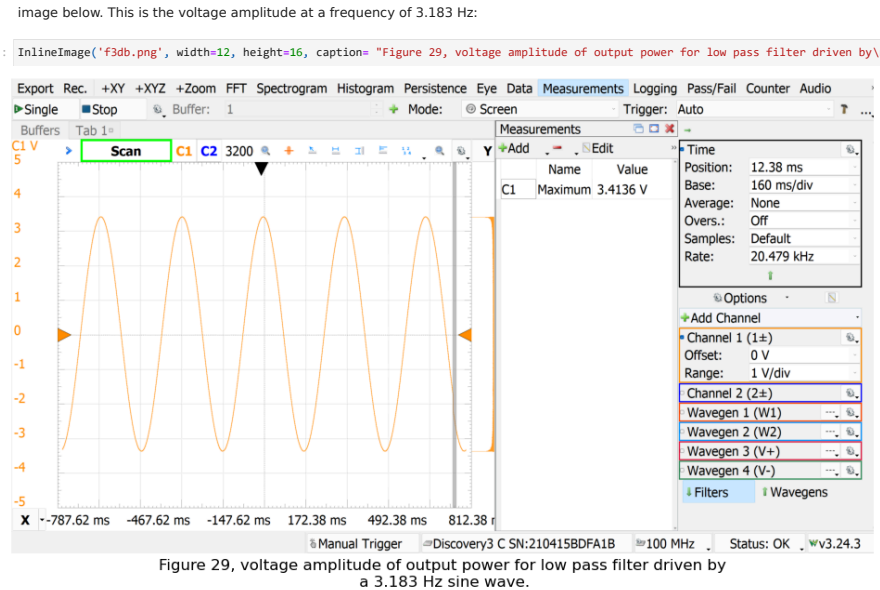

System Monitor
Temperature: 45.00 °C / 113.00 °F
Die: 55.94 °C / 132.69 °F
USB Voltage: 4.834 V
AUX Voltage: 0.031 V
USB Current: 439.9 mA
AUX Current: 0.0 mA
V+: 0.101 V
V-: 0.017 V

Figure 29, voltage output of my computer through usb

As we can see, my computer only outputs 4.834 V. Ok, so lets recalculate. we should be looking for a value of $4.834/\sqrt{2}$, which is 3.418... V. Wow, this matches up with the measurement taken for the voltage amplitude! I have attached an image below. This is the voltage amplitude at a frequency of 3.183 Hz:

```
In [63]: InlineImage('f3db.png', width=12, height=16, caption= "Figure 29, voltage amplitude of output power for low pass filter driven by\n a 3.183 Hz sine wave.", fs=12);
```


Export Rec. +XY +XYZ +Zoom FFT Spectrogram Histogram Persistence Eye Data Measurements Logging Pass/Fail Counter Audio
Single Stop Buffer: 1 Mode: Screen Trigger: Auto
Buffers Tab 1
Scan
C1 C2 3200
X −787.62 ms
-467.62 ms
-147.62 ms
172.38 ms
492.38 ms
812.38
Manual Trigger
Discovery3 C SN:210415BDFA1B
100 MHz
Status: OK
v3.24.3

Measurements: C1 Maximum 3.4136 V

Time Position: 12.38 ms
Base: 160 ms/div
Average: None
Overs.: Off
Samples: Default
Rate: 20.479 kHz
Channel 1 (1x) Offset: 0 V
Range: 1 V/div

Figure 29, voltage amplitude of output power for low pass filter driven by
a 3.183 Hz sine wave.

3.4136. Very close to our expected 3.4182 from my computer's not-quite-5V power source.

This part went quite smoothly as I had most of my troubleshooting issues (sooo much struggling) in the first part measuring C. No setbacks at all here which I am quite happy with!

##2.3 filter design, filter sine wave.

From the file, I know that the frequency of the sine wave is 10kHz. this means that the noise is at a higher frequency, and so we want a low pass filter to filter out the noise. I also know that my filter cannot filter too much; otherwise I will start to lose the signal of the main sine wave. A stronger filter means cleaner wave, but I may lose more voltage amplitude from the waveform I want to clean up. A weaker filter means a messier wave, but I preserve more of the waveform I want to clean; there are tradebacks, so I should aim for a middleground when I am picking my 3dB value.

first, I will find the approx frequency of the wave. Then, I can use this value to approximate the 3dB value. I want to keep the main wave, and then diminish any waves with higher frequency. I can do this by finding an approximate period using the cursors:

```
In [64]: InlineImage('approx.png', width=12, height=16, caption= "Figure 30, using cursors to find the period of the source wave.", fs=12);
```

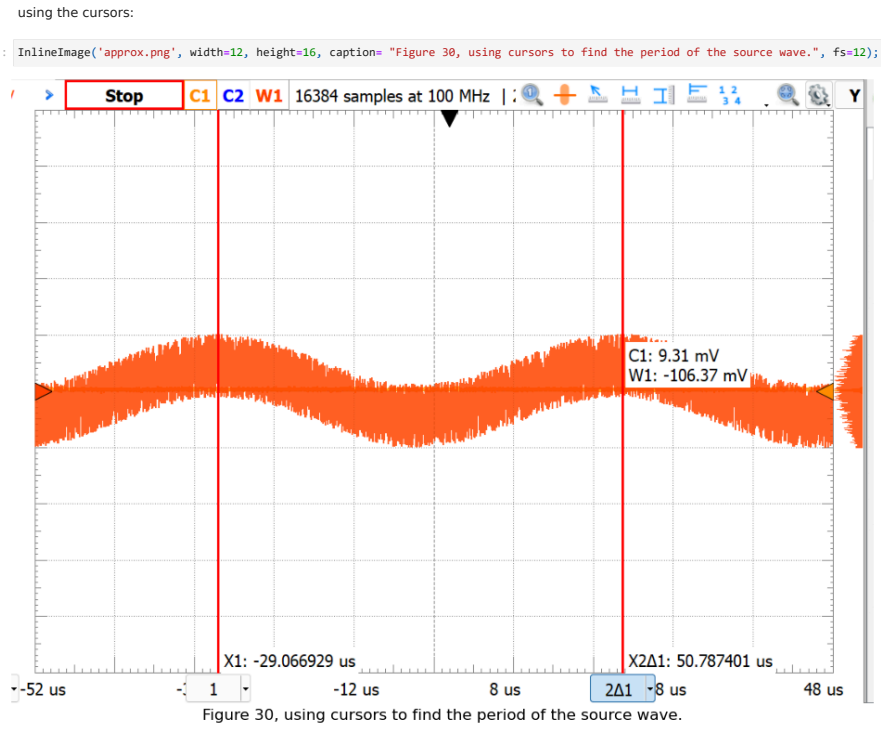

Stop
C1 C2 W1 16384 samples at 100 MHz
C1: 9.31 mV
W1: -106.37 mV
X1: -29.066929 us
X2Δ1: 50.787401 us
−52 us
-12 us
8 us
48 us
1
2Δ1

Figure 30, using cursors to find the period of the source wave.

Using this, I rearranged the equation for the 3dB thing to solve for RC:

$$RC=\frac{1}{2\pi f}=\frac{1}{2\pi 20000}=7.95774x10^{-6}$$

RC should have this value. I tested dividing this by 1nF, and I got a value of 7957 Ohms. so that means I should be using the 1nF capacitor or be in that order of magnitude, and use a resistor close to 5-10 k Ω.

TS by strategically checking and calculating, (with my available resistors and capacitors - 2x 1nF, 2x 1uF, 2x 10k, 1x 5k, 1x 1k resistors), I eventually got 3 possible combinations with a 3dB frequency close to the source wave:

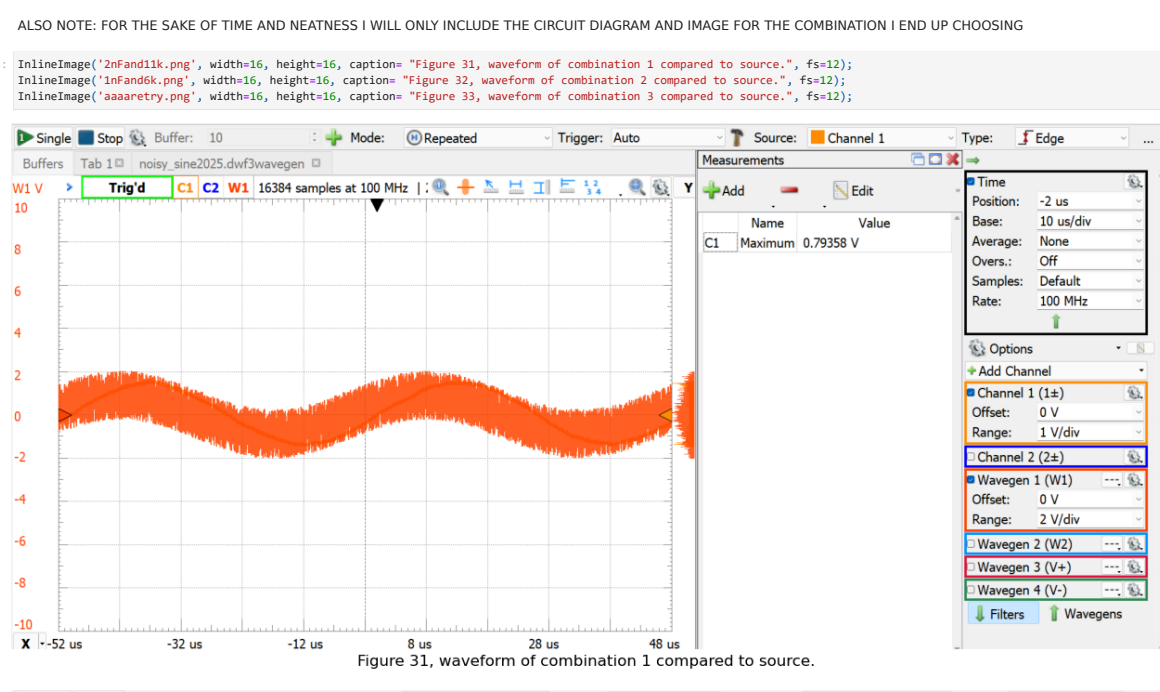
1. 2x 1nF in series, 10 $k\Omega$ + 1 $k\Omega$ resistor in series: $f_{3dB}=28937.26238...Hz$
2. 1x 1nF, 5 $k\Omega$ + 1 $k\Omega$ resistors in series: $f_{3dB}=26525.82385...Hz$
3. 2x 1nF in series, 10 $k\Omega$ + 5 $k\Omega$ resistors in series: $f_{3dB}=21220.65908...Hz$

I will test them to see the waves they result in (see figures below)

NOTE: THE 5K RESISTOR IS THE UNKNOWN ONE FROM THE FIRST PART!!! I think it is okay to use as we found the value. I also ended up confirming with the volt meter prior to this section.

ALSO NOTE: FOR THE SAKE OF TIME AND NEATNESS I WILL ONLY INCLUDE THE CIRCUIT DIAGRAM AND IMAGE FOR THE COMBINATION I END UP CHOOSING

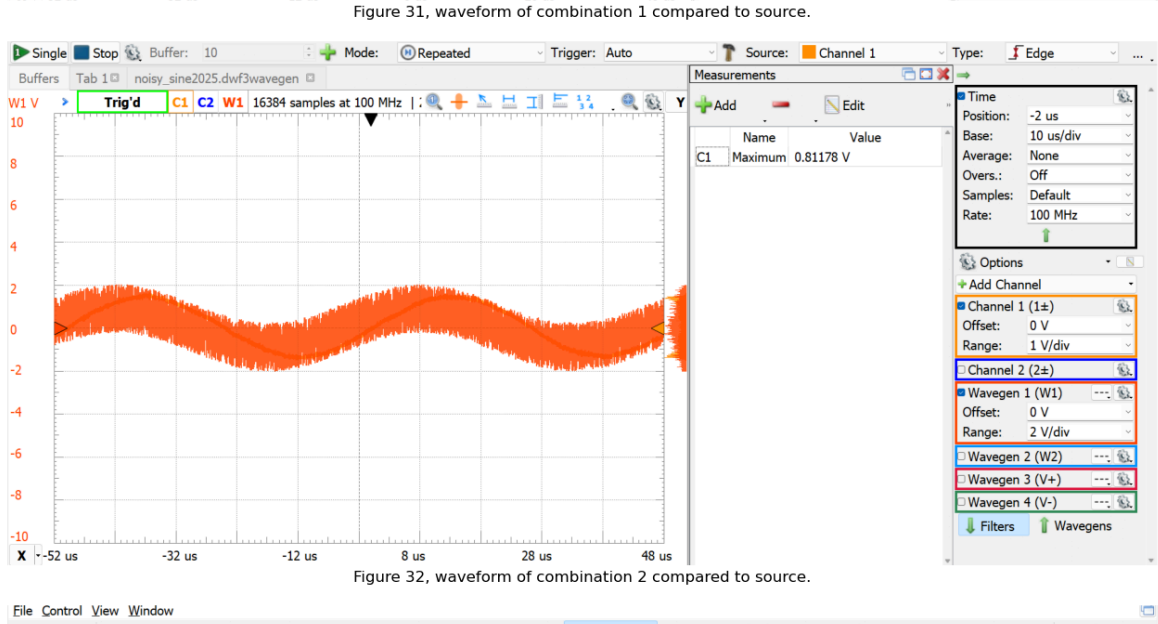
```
In [67]: InlineImage('2nFand11k.png', width=16, height=16, caption= "Figure 31, waveform of combination 1 compared to source.", fs=12);
         InlineImage('1nFand6k.png', width=16, height=16, caption= "Figure 32, waveform of combination 2 compared to source.", fs=12);
         InlineImage('aaaaretry.png', width=16, height=16, caption= "Figure 33, waveform of combination 3 compared to source.", fs=12);
```


Single Stop Buffer: 10 Mode: Repeated Trigger: Auto Source: Channel 1 Type: Edge
Buffers Tab 1 noisy_sine2025.dwf3wavegen
Trig'd
C1 C2 W1 16384 samples at 100 MHz
X −52 us
-32 us
-12 us
8 us
28 us
48 us

Measurements: C1 Maximum 0.79358 V

Time Position: -2 us
Base: 10 us/div
Average: None
Overs.: Off
Samples: Default
Rate: 100 MHz
Channel 1 (1x) Offset: 0 V
Range: 1 V/div
Wavegen 1 (W1) Offset: 0 V
Range: 2 V/div

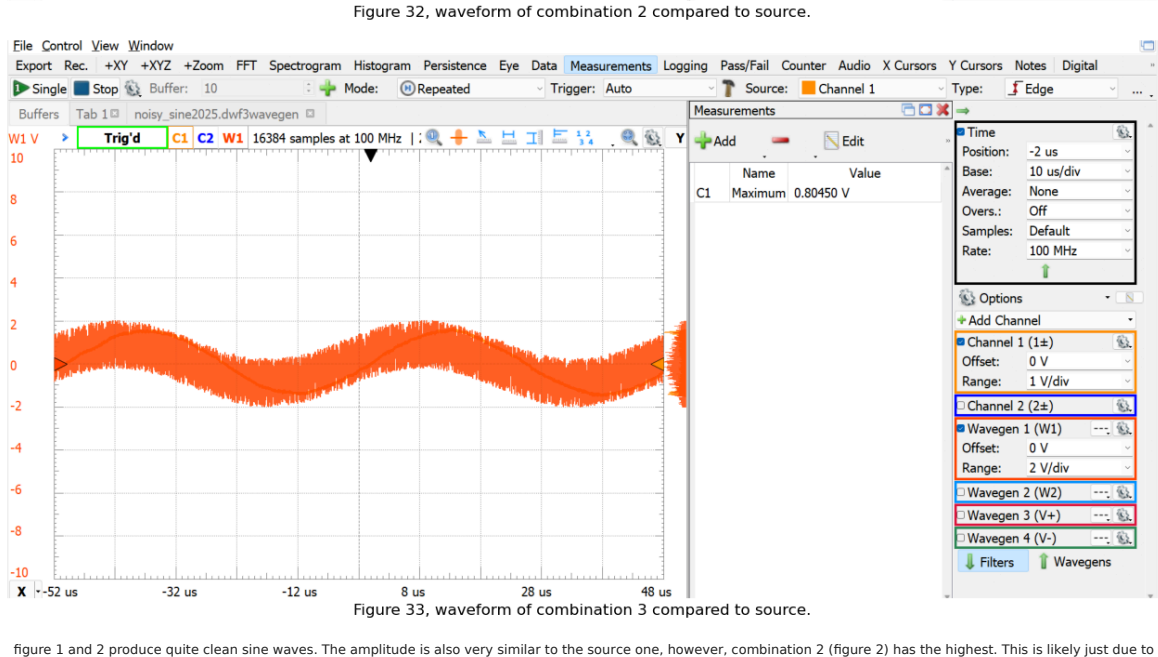
Figure 31, waveform of combination 1 compared to source.


Single Stop Buffer: 10 Mode: Repeated Trigger: Auto Source: Channel 1 Type: Edge
Buffers Tab 1 noisy_sine2025.dwf3wavegen
Trig'd
C1 C2 W1 16384 samples at 100 MHz
X −52 us
-32 us
-12 us
8 us
28 us
48 us

Measurements: C1 Maximum 0.81178 V

Time Position: -2 us
Base: 10 us/div
Average: None
Overs.: Off
Samples: Default
Rate: 100 MHz
Channel 1 (1x) Offset: 0 V
Range: 1 V/div
Wavegen 1 (W1) Offset: 0 V
Range: 2 V/div

Figure 32, waveform of combination 2 compared to source.


File Control View Window
Export Rec. +XY +XYZ +Zoom FFT Spectrogram Histogram Persistence Eye Data Measurements Logging Pass/Fail Counter Audio X Cursors Y Cursors Notes Digital
Single Stop Buffer: 10 Mode: Repeated Trigger: Auto Source: Channel 1 Type: Edge
Buffers Tab 1 noisy_sine2025.dwf3wavegen
Trig'd
C1 C2 W1 16384 samples at 100 MHz
X −52 us
-32 us
-12 us
8 us
28 us
48 us

Measurements: C1 Maximum 0.80450 V

Time Position: -2 us
Base: 10 us/div
Average: None
Overs.: Off
Samples: Default
Rate: 100 MHz
Channel 1 (1x) Offset: 0 V
Range: 1 V/div
Wavegen 1 (W1) Offset: 0 V
Range: 2 V/div

Figure 33, waveform of combination 3 compared to source.

figure 1 and 2 produce quite clean sine waves. The amplitude is also very similar to the source one, however, combination 2 (figure 2) has the highest. This is likely just due to the fluctuation of the readings, and related to the uncertainty. However, I will take combination 2 to be the correct one moving forward. Figure 3 also has a high voltage reading. it seems all of them work. However, just because combination 2 has the highest, I will take that setup going forward.

Below is my circuit diagram for combination 2, as well as a labelled image. I also show the filtered end result waveform by itself.

```
In [68]: InlineImage('yayayayaya.png', width=16, height=16, caption= "Figure 34, circuit diagram of combination 2 low pass filter for 3dB frequency of 26525 Hz", fs=12);
         InlineImage('yayayaya.png', width=16, height=16, caption= "Figure 35, circuit image of combination 2 low pass filter for 3dB frequency of 26525 Hz", fs=12);
         InlineImage('final.png', width=16, height=16, caption= "Figure 36, final waveform of combination 2.", fs=12);
```

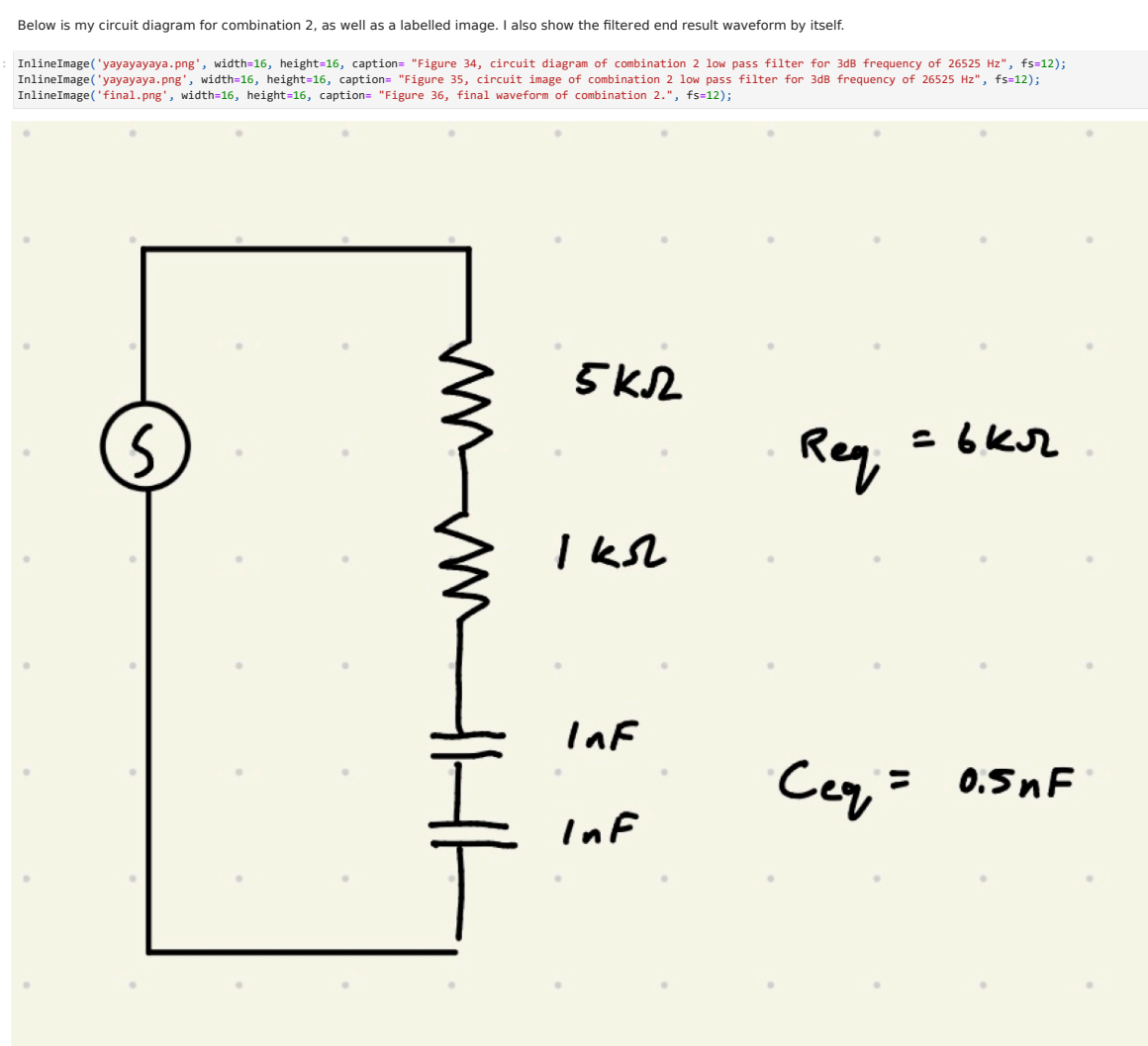

5 kΩ
1 kΩ
1nF
1nF
Req = 6kΩ
Ceq = 0.5nF

Figure 34, circuit diagram of combination 2 low pass filter for 3dB frequency of 26525 Hz

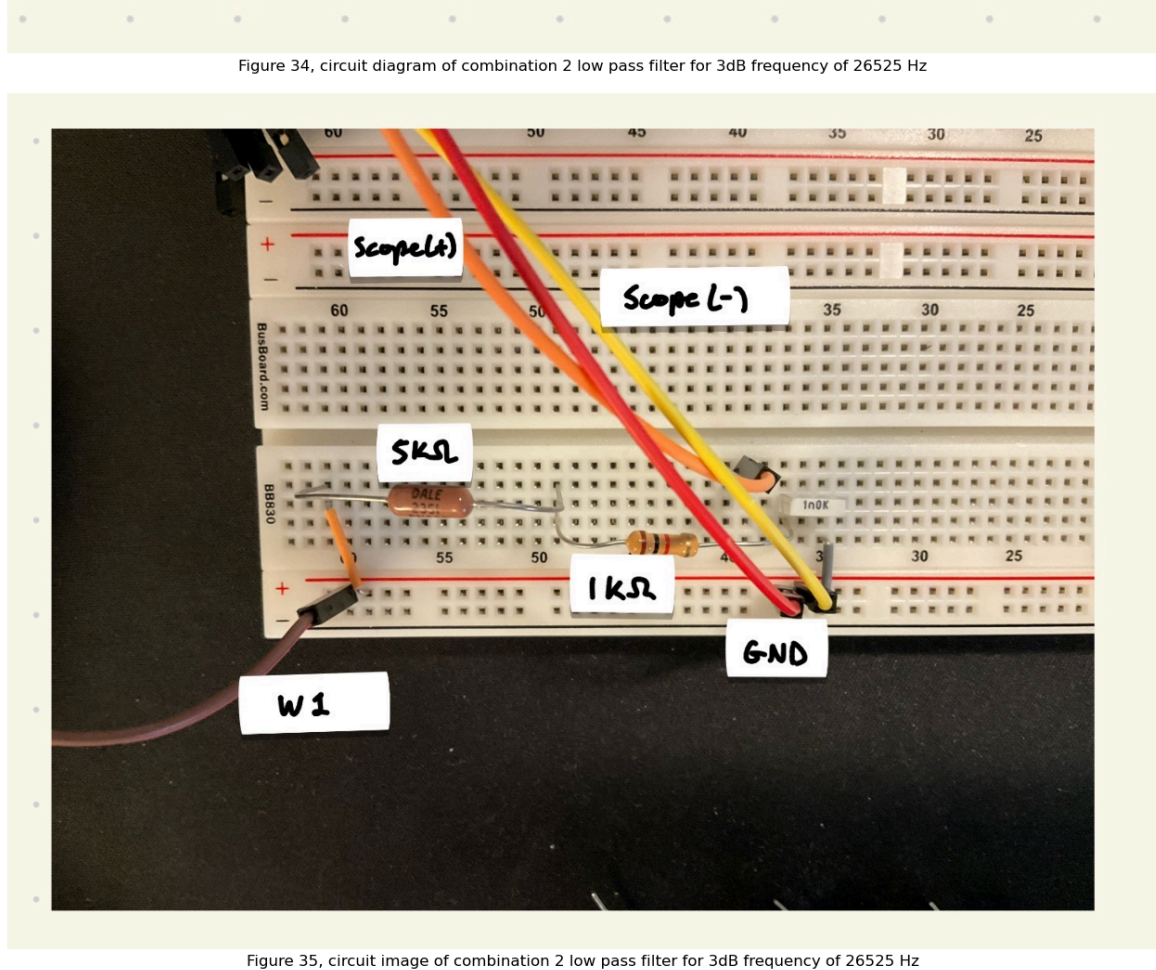

Scope(+)
Scope(-)
5kΩ
1kΩ
GND
W1

Figure 35, circuit image of combination 2 low pass filter for 3dB frequency of 26525 Hz

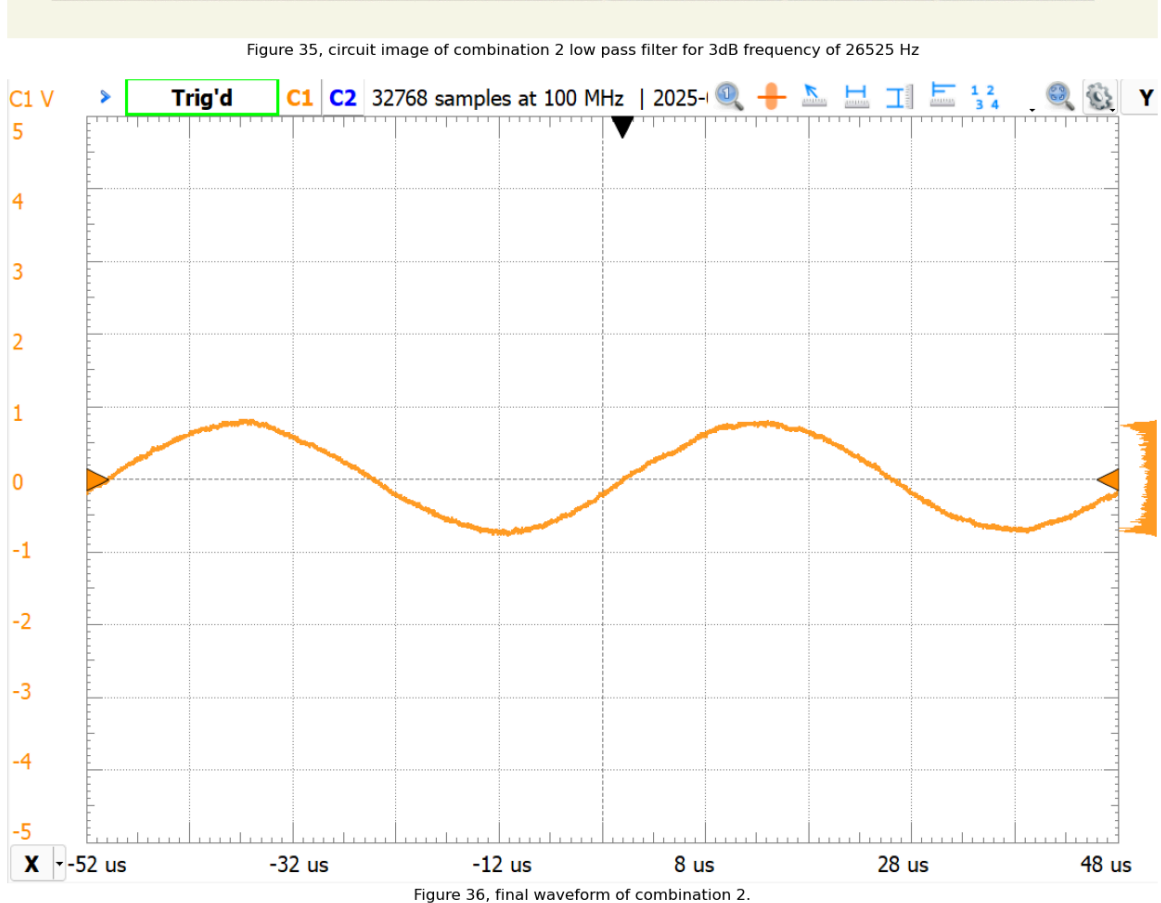

C1 V
Trig'd
C1 C2 32768 samples at 100 MHz | 2025-
5
4
3
2
1
0
-1
-2
-3
-4
-5
X −52 us
-32 us
-12 us
8 us
28 us
48 us

Figure 36, final waveform of combination 2.

#3 conclusion

In this lab, we measured the values of an unknown resistor using a circuit involving a known resistor and power source. We were able to calculate the current in the circuit, and then obtain a value for the resistance with uncertainty. Then, we measured the value of an ukown capacitor. This was a challenge that had multiple possible methods, all of which involved the time constant. The time constant proves to be a powerful classification tool for dissecting circuits and analyzing them in the time domains.

After this, we measured the 3dB frequency, a frequency for a given circuit in which the output power is $1/\sqrt{2}$ times the maximum. This proved useful in the step afterwards, which involved analysis of this value (in an estimate) to build a low pass filter for a given sine wave. With a bit of strategic calculation and process of elimination, I was able to successfully utilize different parts of the lab to construct a working low pass filter.

Key takeaways:

- there is a difference between the power source and the wave source
- in a given circuit with a 3dB value, increasing the frequency of the ac source past that value will diminish the amplitude of the source significantly. (basically how a low pass filter works)
- we can set the 3dB value by changing C or R as we like. We can also do this in accordance to a wave, by analyzing the target wave's period.
- my computer does not supply 5 volts exactly.
- despite the slope method being quite precise, the other method can save lots of time if that is of interest
- we can probably use these learnings to create a high pass filter to only keep the noise too (by changing the order of resistors and capacitors)
- building a circuit with a 3dB value too close to the target wave's frequency can result in the target wave being filtered out too/losing amplitude; we must be careful as to which value we build into the filter.

```
In [2]: # comments in Python start with a "#" and will not be executed - well commented code is easier for graders to understand and give partial marks to ;)
        # Importing modules that will be necessary for the tutorial
        import numpy as np # numpy is a general math/array manipulation library that is almost always used in any scientific/engineering notebook
        import matplotlib.image as mplimg # matplotlib.image is used for displaying images in python
        import matplotlib.pyplot as plt # matplotlib.pyplot is used for general plotting and is very similar to matlab syntax for plotting
        import pandas as pd # pandas is a data manipulation library, we will use it to import data from excel files
        print('All necessary modules imported!')
```

```
All necessary modules imported!
```

```
In [4]: def InlineImage(img, width=8, height=8, caption=None, fs=24):
            # Author: BG
            # Renders an image with a caption underneath using matplotlib
            # Requires matplotlib.pyplot to be imported as plt and matplotlib.img to be imported as mplimg
            # img - string, the location of an image file
            # width, height - numbers, the dimensions of the figure the image is displayed in. Original aspect ratio of the image will be preserved
            # caption - string, the caption to be displayed beneath the image
            # fs - number, the font size of the caption
            fig,ax = plt.subplots(figsize=(width,height))
            ax.imshow(mplimg.imread(img))
            ax.set_xlabel(caption,fontsize=fs)
            ax.tick_params(
                which='both',
                bottom=False,
                top=False,
        left=False,
        labelleft=False,
        labelbottom=False)
    ax.spines["top"].set_visible(False)
    ax.spines["left"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.spines["bottom"].set_visible(False)
```

```
In [ ]:
```